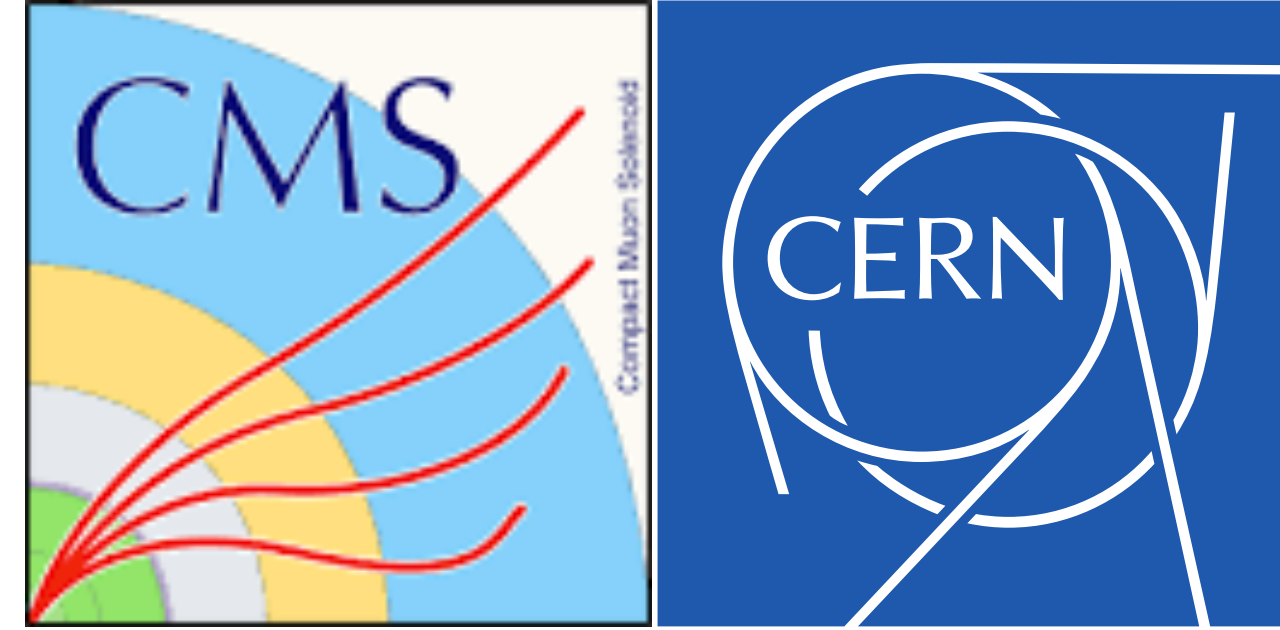




University
of Split



DATA ANALYSIS

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- 1) Introduction to Data Analysis
- 2) Probability density functions and Monte Carlo methods
- 3) Parameter estimation
- 4) Confidence intervals
- 5) Hypothesis testing and p-value

PROBABILITY DENSITY FUNCTIONS

- Let x be a possible outcome of an observation and can take any value from a continuous range
- We write $f(x;\theta)dx$ as the probability that the measurement's outcome lies between x and $x + dx$
- The function **$f(\mathbf{x};\boldsymbol{\theta})d\mathbf{x}$** is called the **probability density function (PDF)**
 - And may depend on one or more parameters θ
- If $f(x;\theta)$ can take only discrete values then $f(x;\theta)$ is itself a probability
- The p.d.f. is always normalised to a unit area (unit sum, if discrete)
- Both **$\mathbf{x}$** and **$\boldsymbol{\theta}$** may have multiple components and are then written as vectors

$$P(x \in [x, x + dx] | \theta) = f(x; \theta)dx$$

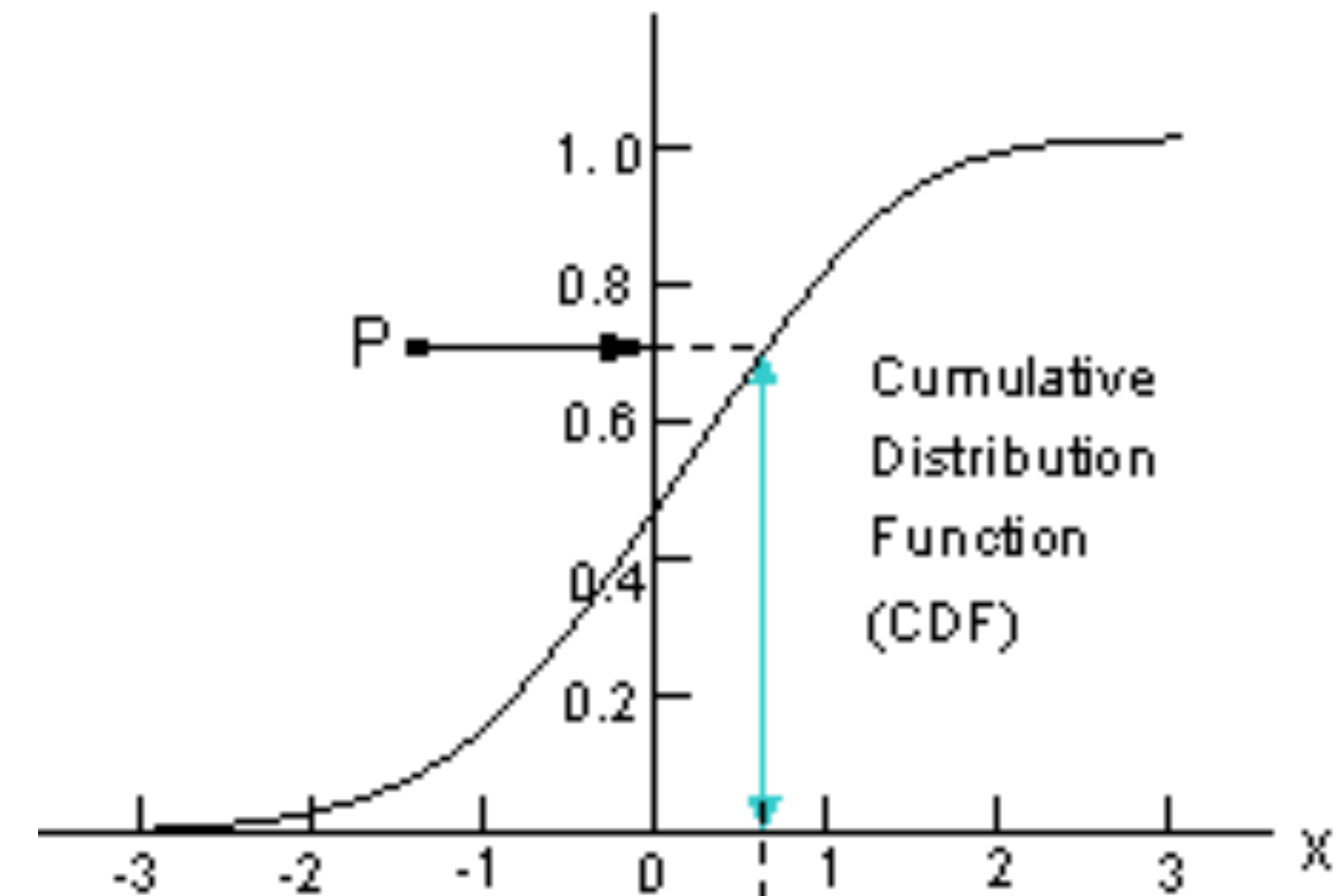
$$\int_{-\infty}^{\infty} f(x; \theta)dx = 1$$

● Cumulative distribution function, CDF

- for every real number Y , the CDF of Y is equal to the probability that the random variable x takes a value less or equal to Y

$$F(Y) = P(x \leq Y) = \int_{x_{\min}}^Y f(x)dx$$

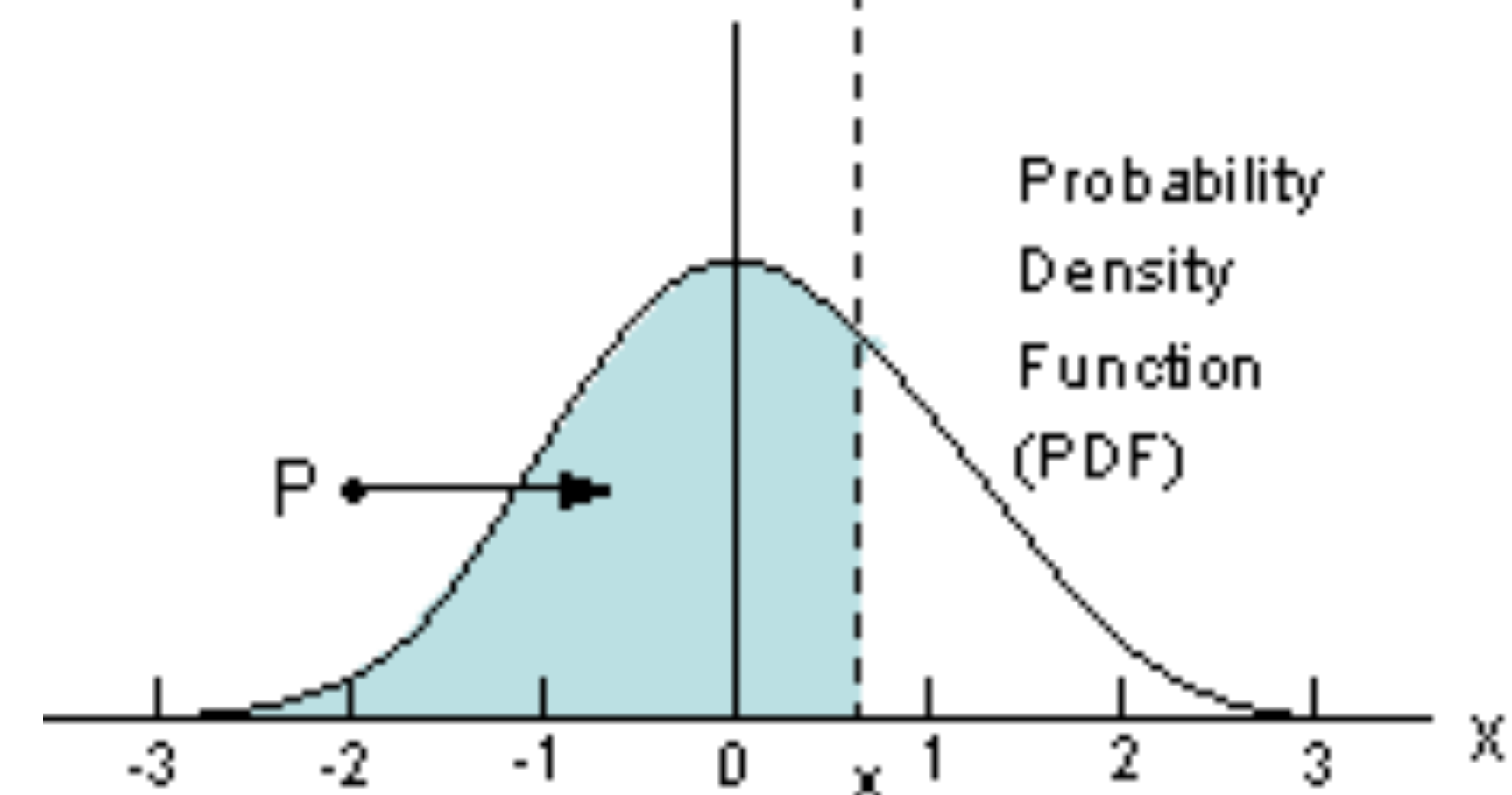
- If x restricted to $x_{\min} < x < x_{\max}$ then $F(x_{\min}) = 0$, $F(x_{\max}) = 1$
- $F(x)$ is a monotonic function of x



● Marginal density function

- is the projection of multidimensional density
- Example: if $f(x,y)$ is two-dimensional PDF the marginal density $g(x)$ is

$$g(x) = \int_{y_{\min}}^{y_{\max}} f(x, y)dy$$



- Probability density function (PDF) = $f(x)dx$
- Expectation:
 - Expectation of any random function $g(x)$: $E(g) = \int g(x)f(x)dx$
 - Expectation of x is the **mean**: $\mu = E(x) = \int xf(x)dx$
- **Variance**: $V(x) = \sigma^2 = E[(x - \mu)^2] = \int (x - \mu)^2 f(x)dx$
- $E(x)$ is usually a measure of the **location** of the distribution
- $V(x)$ is usually a measure of the **spread** of the distribution

- Probability for r successes is given by the Binomial distribution:

$$P(r; p, N) = \binom{N}{r} p^r (1 - p)^{N-r}$$

- $P(r; N, p)$ is a probability of finding exactly r successes in N trials, when probability of success in each single trial is a constant, p

- Properties of the Binomial distribution:

- Mean: $\langle r \rangle = E(r) = Np$

- Variance: $V(r) = Np(1 - p)$

● Usage example 1:

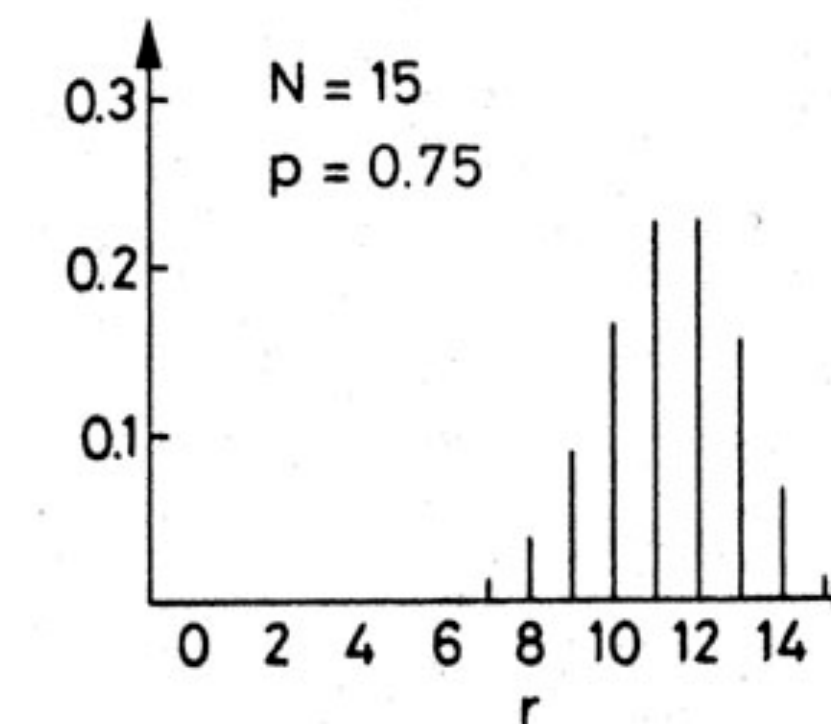
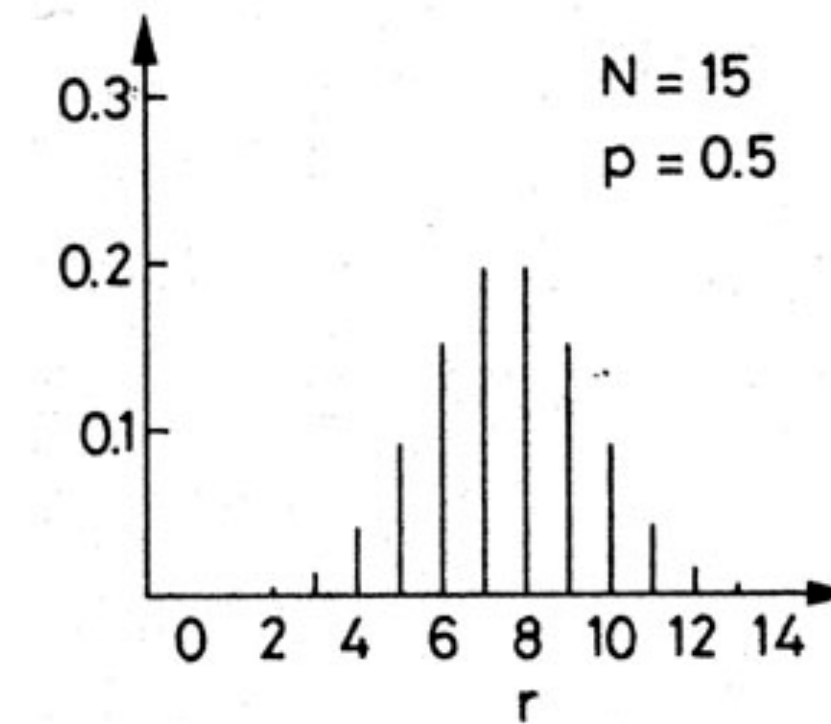
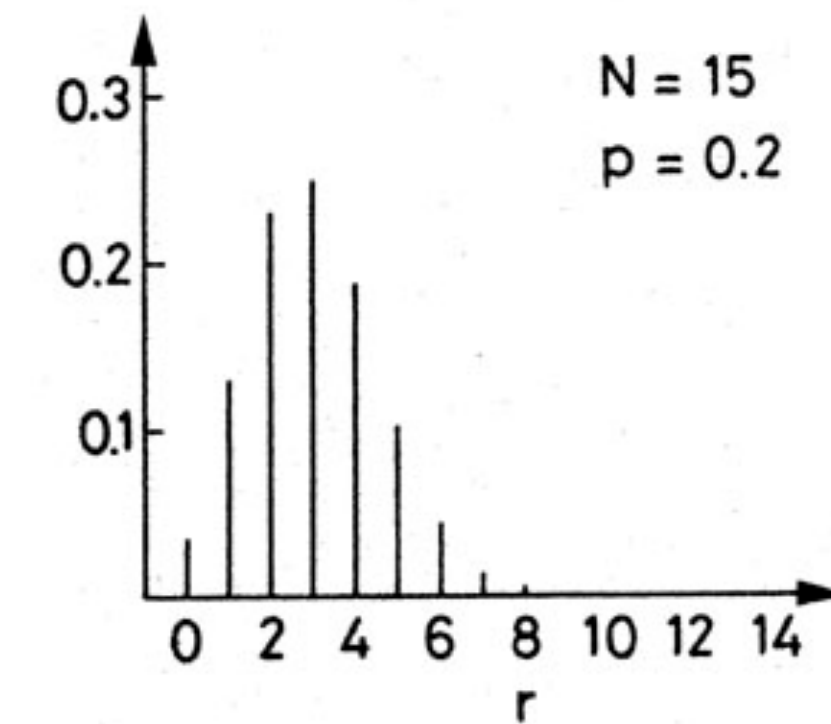
- Probability for a Z boson to decay to two electrons is 3%. What is the probability to find exactly 5 $Z \rightarrow ee$ events out of 80 Z decays?

$$P(5; 0.03, 80) = \binom{80}{5} 0.03^5 (1 - 0.03)^{80-5} = 6 \%$$

● Usage example 2:

- If you flip a biased coin that has a 99% probability of landing on heads, what is the probability to get heads all 6 times from 6 throws?

$$P(6; 0.99, 6) = \binom{6}{6} 0.99^6 (1 - 0.99)^{6-6} = 94.15 \%$$



FROM BINOMIAL TO POISSON

- λ events expected to occur in average during some time interval
- split interval in n very small divisions: chance of getting two events in one section can be discounted
- probability that a given section contains an event: λ/n
- use Binomial formula to calculate the probability to see r events:

$$P(r; \lambda/n, n) = \binom{n}{r} \frac{\lambda^r}{n^r} \left(1 - \frac{\lambda}{n}\right)^{n-r} = \frac{n!}{r!(n-r)!} \frac{\lambda^r}{n^r} \left(1 - \frac{\lambda}{n}\right)^{n-r}$$

● For $n \rightarrow \infty$: $\left(1 - \frac{\lambda}{n}\right)^{n-r} \rightarrow \left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$

$$\frac{n!}{r!(n-r)!} = \frac{n(n-1)\cdots(n-r+1) \cdot \cancel{(n-r)!}}{r! \cancel{(n-r)!}} \rightarrow \frac{n^r}{r!}$$

- Probability of a number of events occurring in a fixed period of time if these events occur with a known average rate λ and independently of the time since the last event:

$$P(r, \lambda) = \frac{e^{-\lambda} \lambda^r}{r!}$$

- Compared to Binomial distribution still has particular (discrete) outcomes, but number of trials is unknown

- Properties of the Poisson distribution:

- Mean: $\langle r \rangle = E(r) = \lambda$

- Variance: $V(r) = \lambda$

BONUS PROBLEM - 2

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Some rules to follow:

1. In every lecture there will be one bonus problem presented
2. If you have good knowledge in stats and everything I am presenting is known to you feel free to start working on the problem now!
3. Otherwise, work on the problem after the lectures.
4. Solutions won't be provided, you have to come and talk to me to check if your answer is correct or if you need hints!
5. Google/AI assistance is not allowed. These are problems that I want you to think about on your own

Estimate the probability for a pilot to die in an airplane crash during their career.

- Usage example 1:

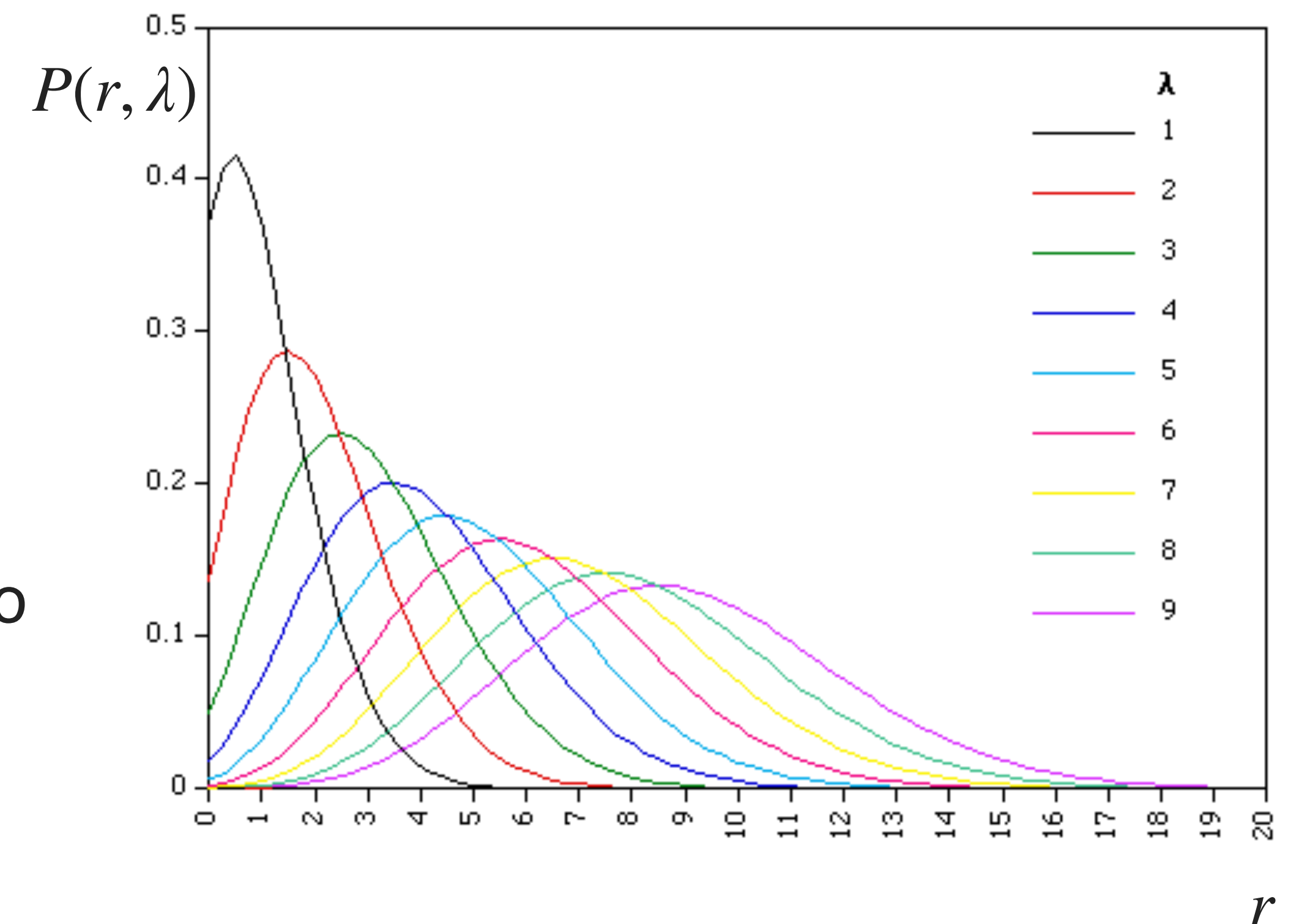
- FIFA reports that the average number of goals in a World Cup soccer match is approximately 2.5. What is the probability to have 5 goals in a match?

$$P(5, 2.5) = \frac{e^{-2.5} 2.5^5}{5!} = 6.7 \%$$

- Usage example 2:

- If expect to detect one extremely high energy gamma ray every year, what is the probability not to detect any in a year?

$$P(0, 1) = \frac{e^{-1} 1^0}{0!} = 36.8 \%$$



● A student is trying to hitch a lift. Cars pass at random intervals at an average rate of 1 per minute. The probability of a car giving a lift is 1%. What is the probability that the student will still be waiting:

(1) after 60 cars have passed?

(2) after 1 hour?

$$(1) \quad P(60; 0.99, 60) = \binom{60}{60} 0.99^{60} (1 - 0.99)^{60-60} = 54.71 \%$$

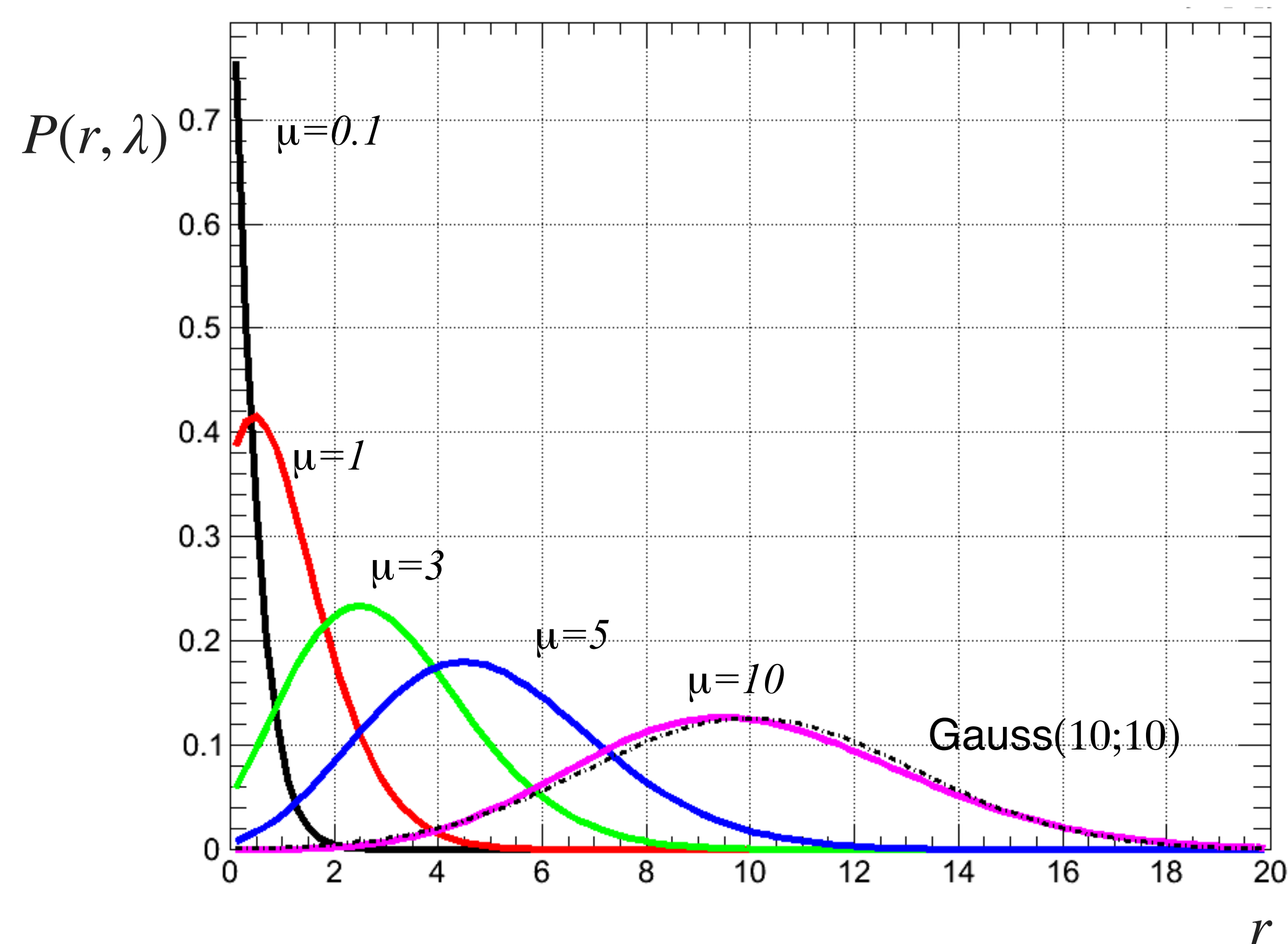
$$(2) \quad P(0, 0.6) = \frac{e^{-0.6} 0.6^0}{0!} = 54.88 \%$$

FROM POISSON TO NORMAL DISTRIBUTION

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- For a large λ Poisson distribution converges towards a Gaussian distribution

$$P(r, \lambda) = \frac{e^{-\lambda} \lambda^r}{r!} \xrightarrow{\text{large } \lambda} \text{Gauss}(r; \mu \equiv \lambda)$$



- The most important distribution in statistics because of the Central Limit Theorem:

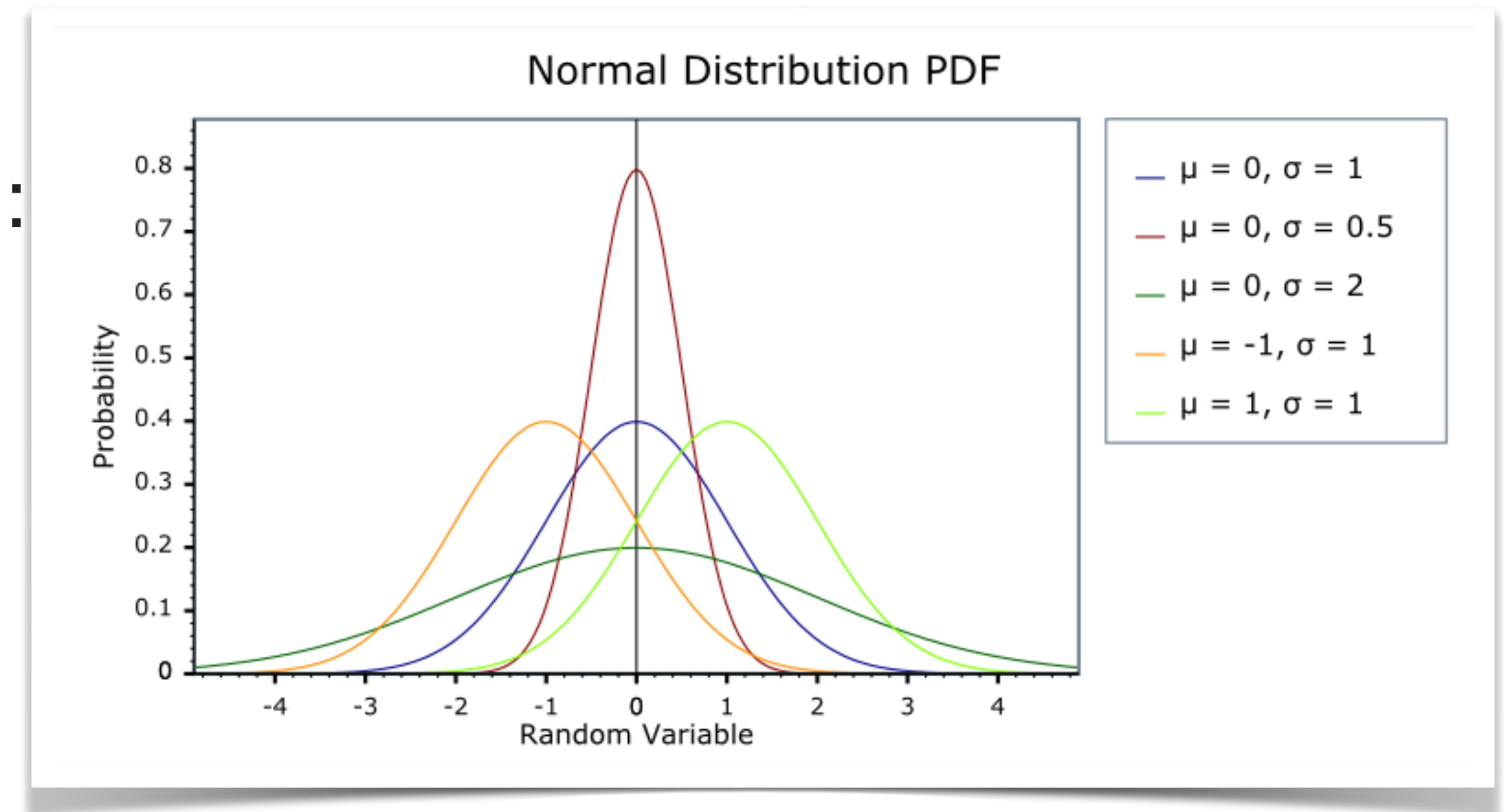
$$N(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- $N(0,1)$ is called standard Normal density

- Properties of the Gaussian distribution:

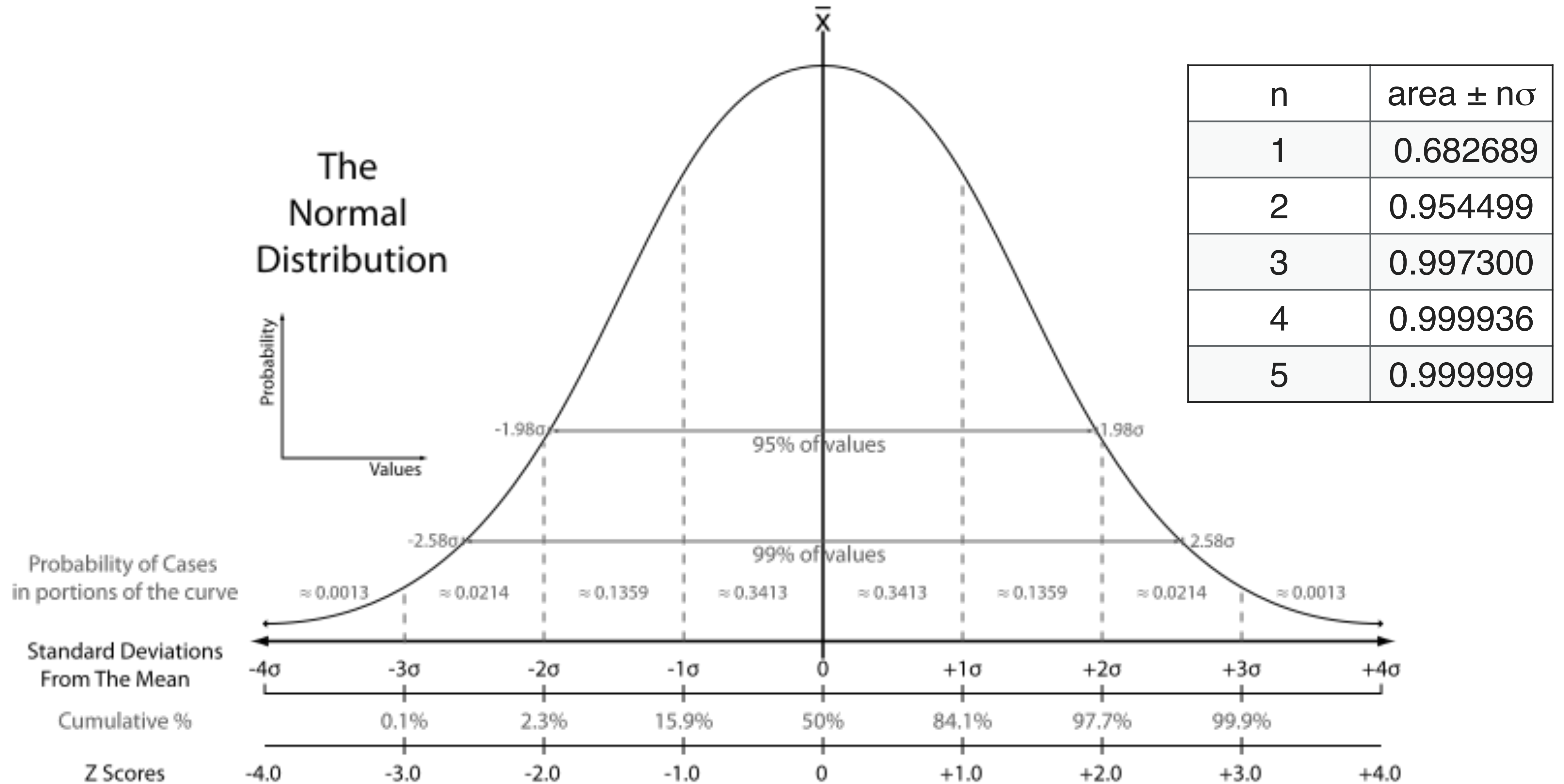
- Mean: $\langle r \rangle = E(r) = \mu$

- Variance: $V(r) = \sigma^2$

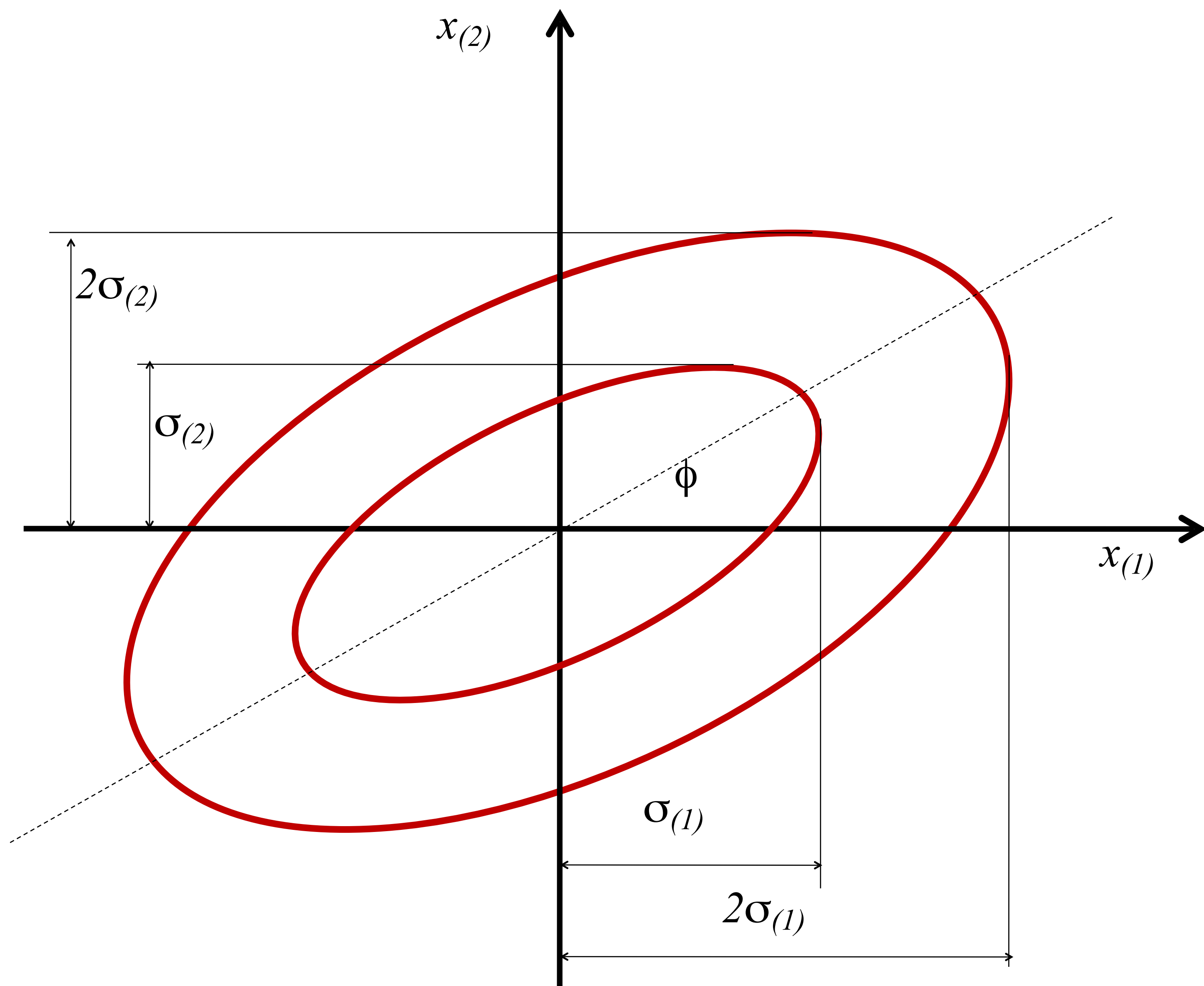


NORMAL DISTRIBUTION PROPERTIES

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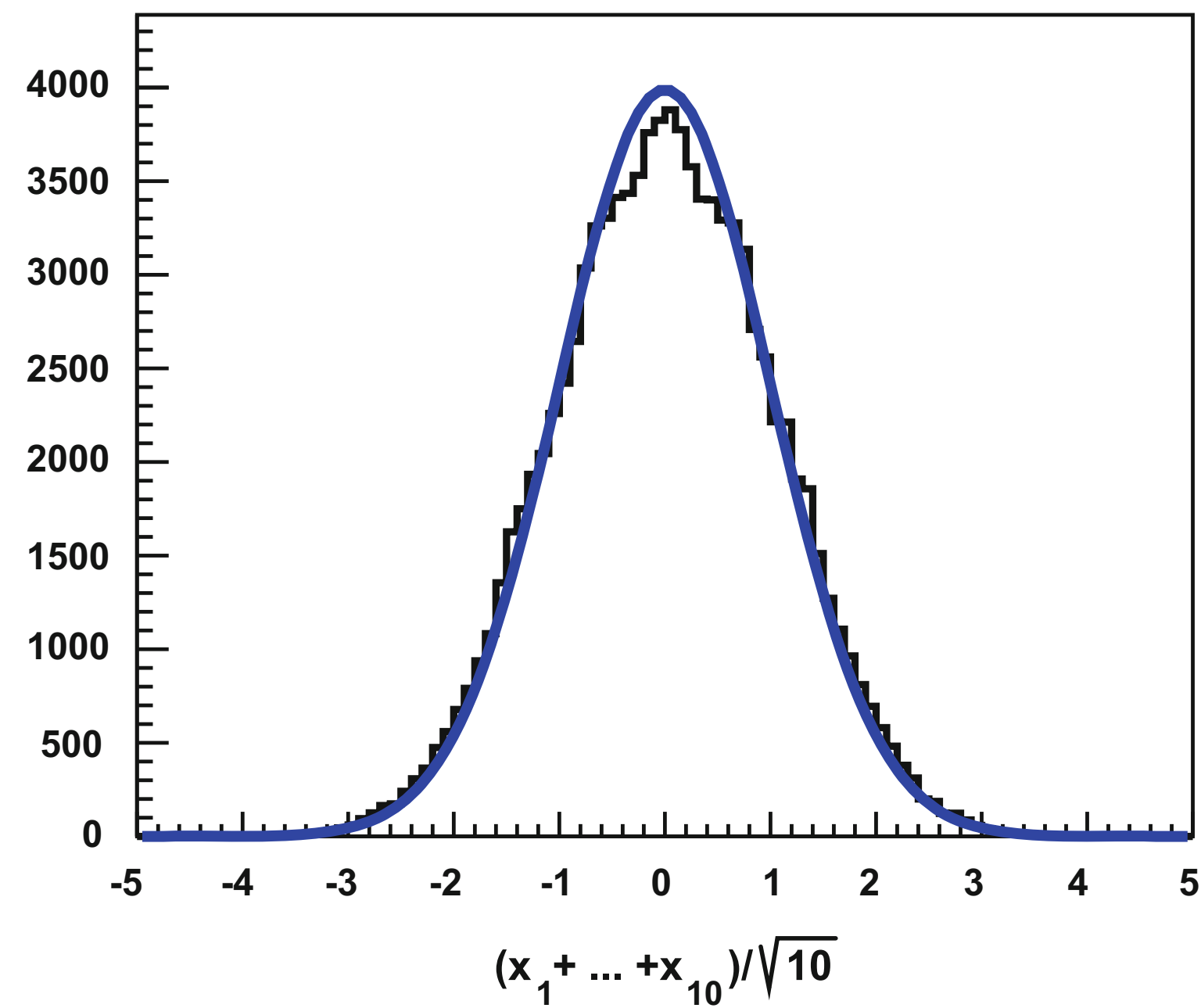
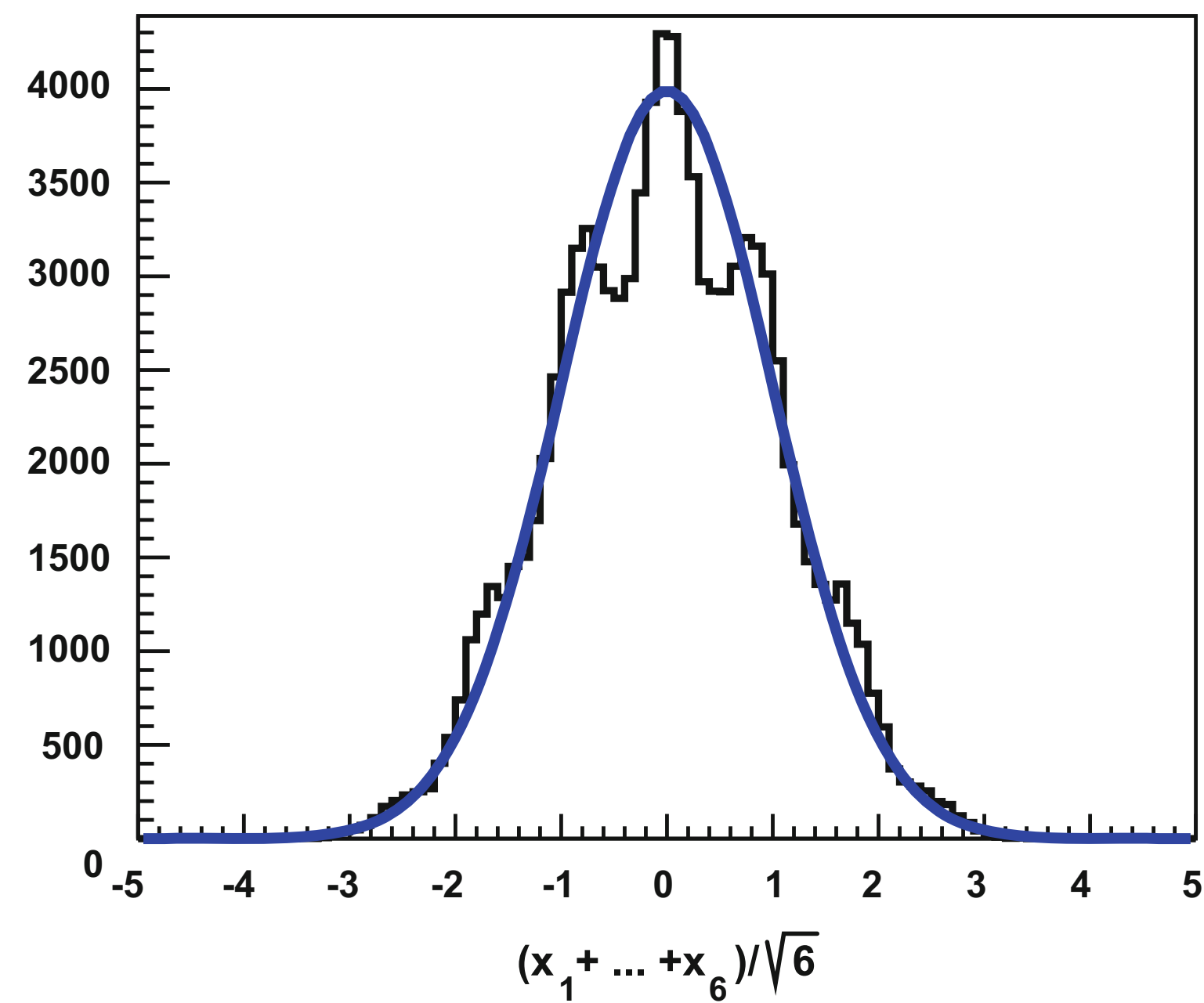
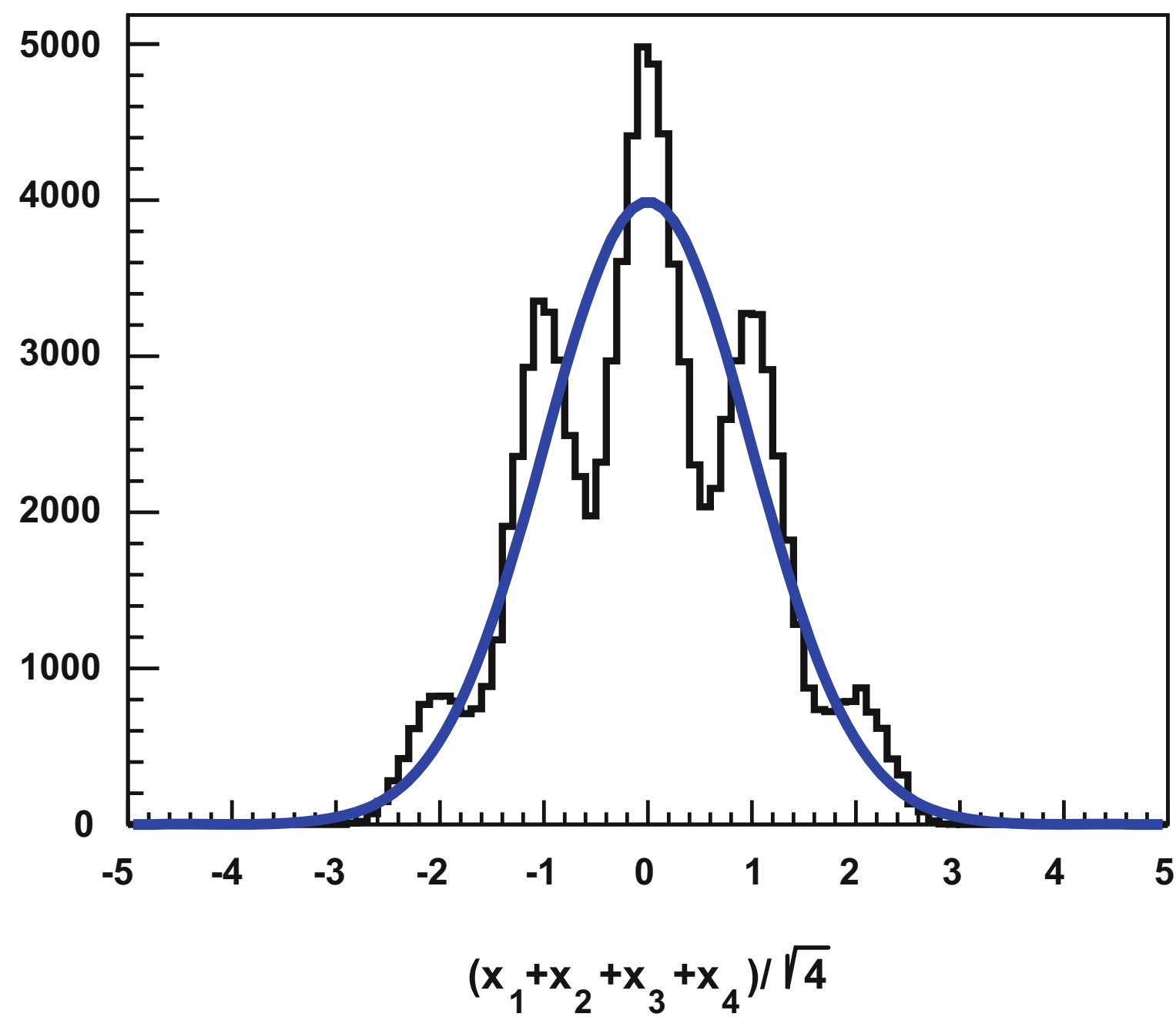
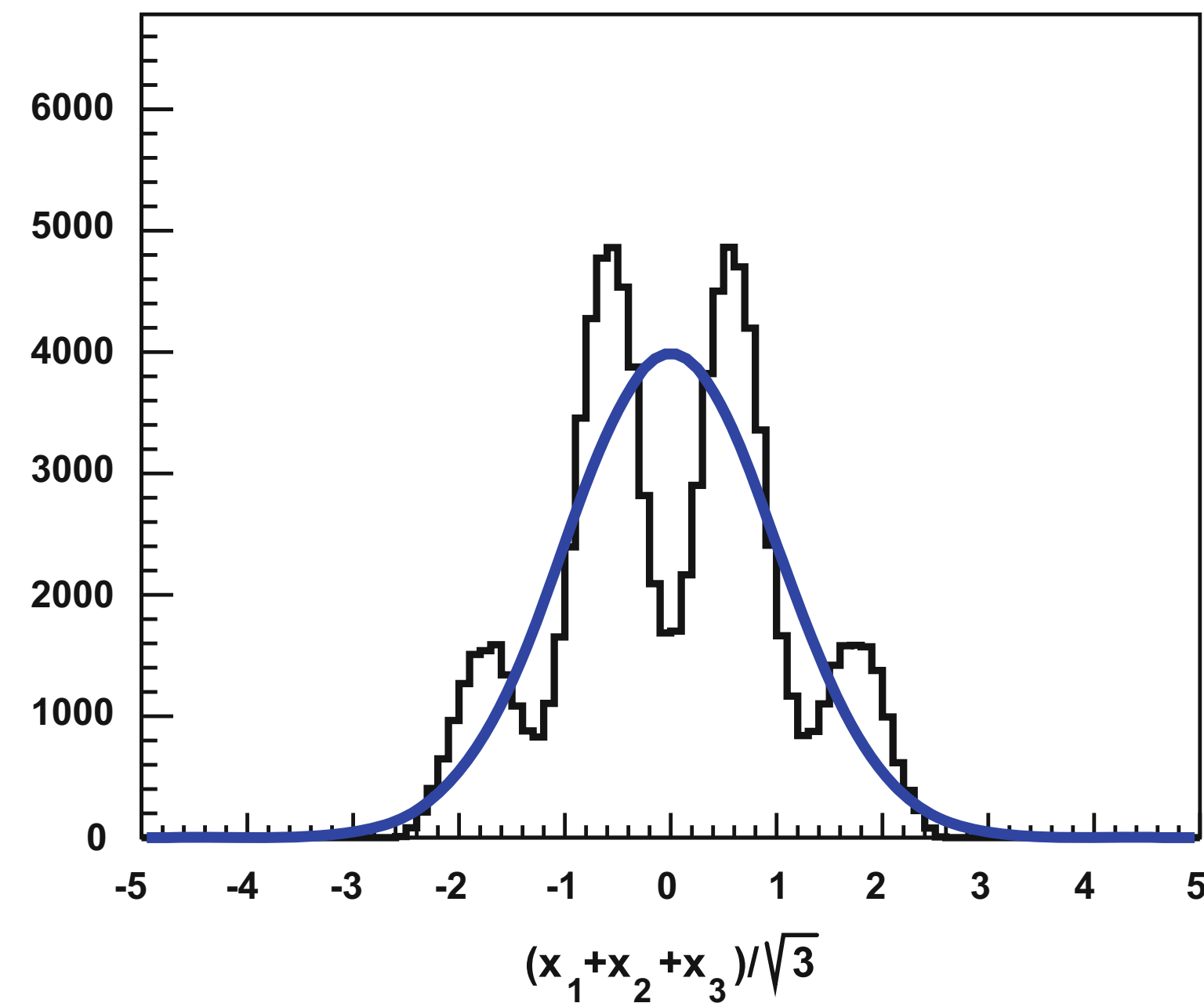
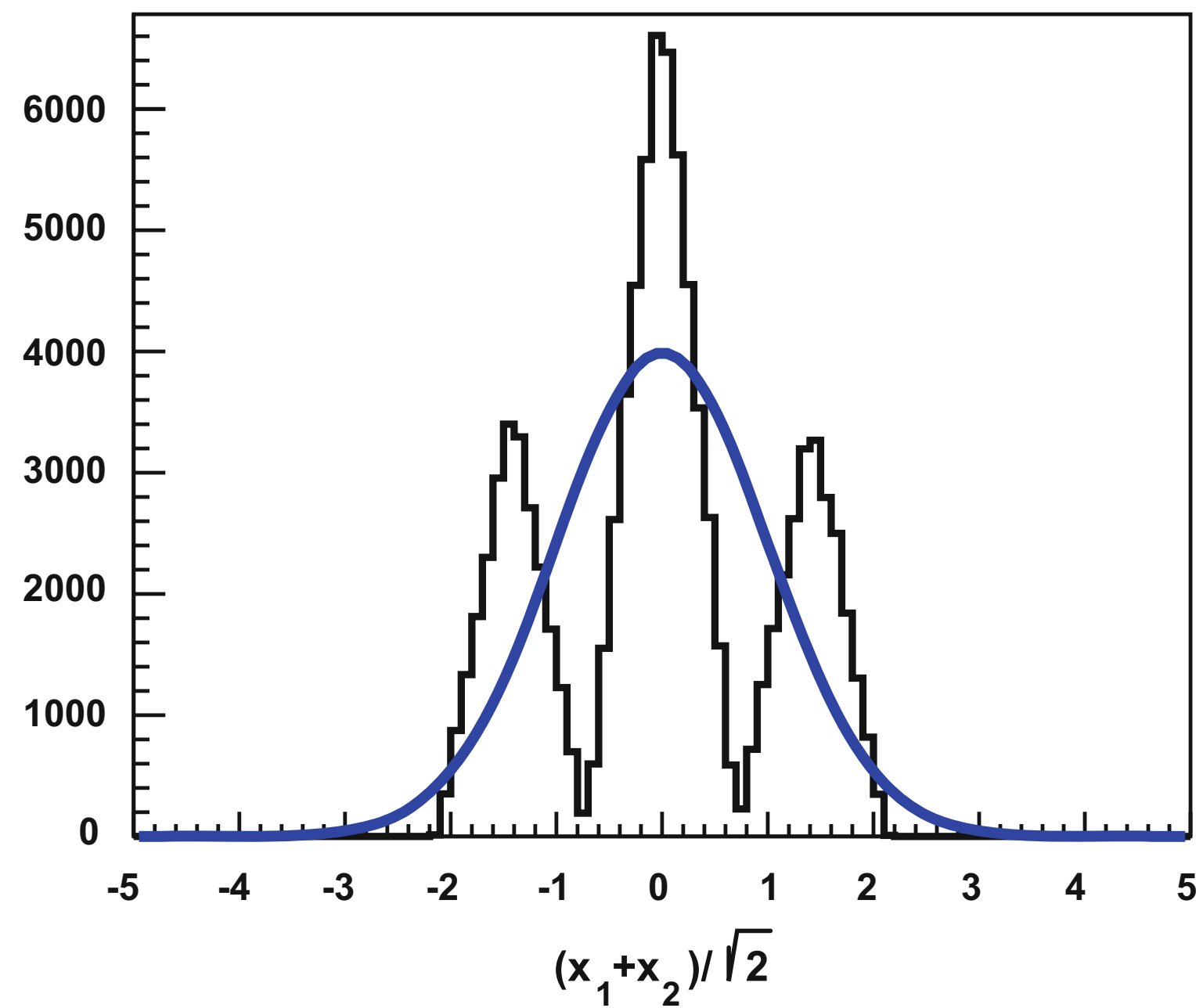
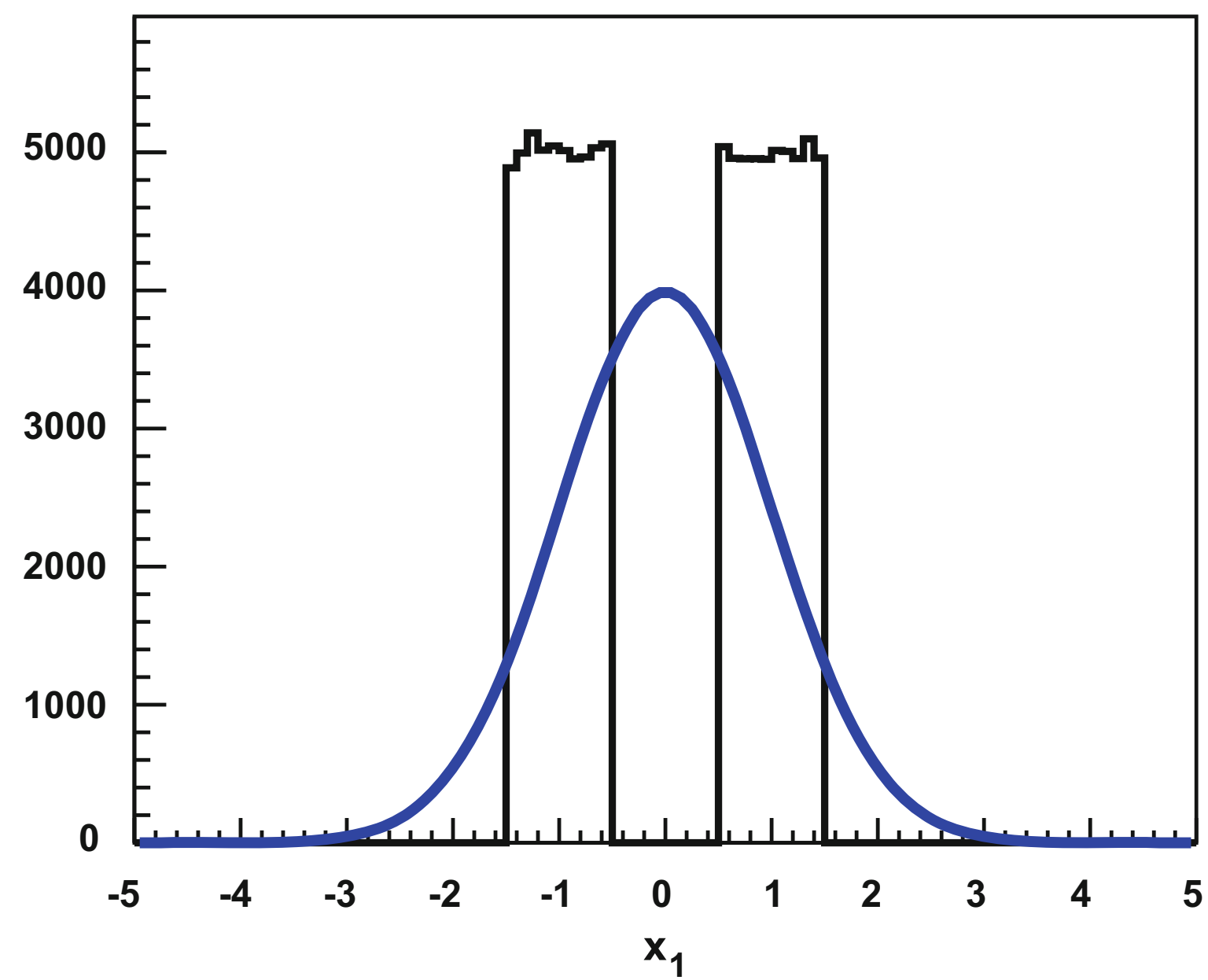
2D GAUSSIAN



n	P_{1D}	P_{2D}
1σ	0.6827	0.3934
2σ	0.9545	0.8647
3σ	0.9973	0.9889
1.515σ		0.6827
2.486σ		0.9545
3.439σ		0.9973

● Central limit theorem:

- If we have a set of N independent variables x_i , each from a distribution with mean μ_i and variance σ_i^2 , then the distribution of the sum $X = \sum x_i$
 - has a mean $\langle X \rangle = \sum \mu_i$,
 - has a variance $V(X) = \sum \sigma_i^2$,
 - becomes Gaussian as $N \rightarrow \infty$.
- Therefore, no matter what the distributions of original variables may have been, their sum will be Gaussian in a large N limit
- Example:
 - measurements errors
 - human heights are well described by a Gaussian distribution, as many other anatomical measurements, as these are due to the combined effects of many genetic and environmental factors
 - student test scores



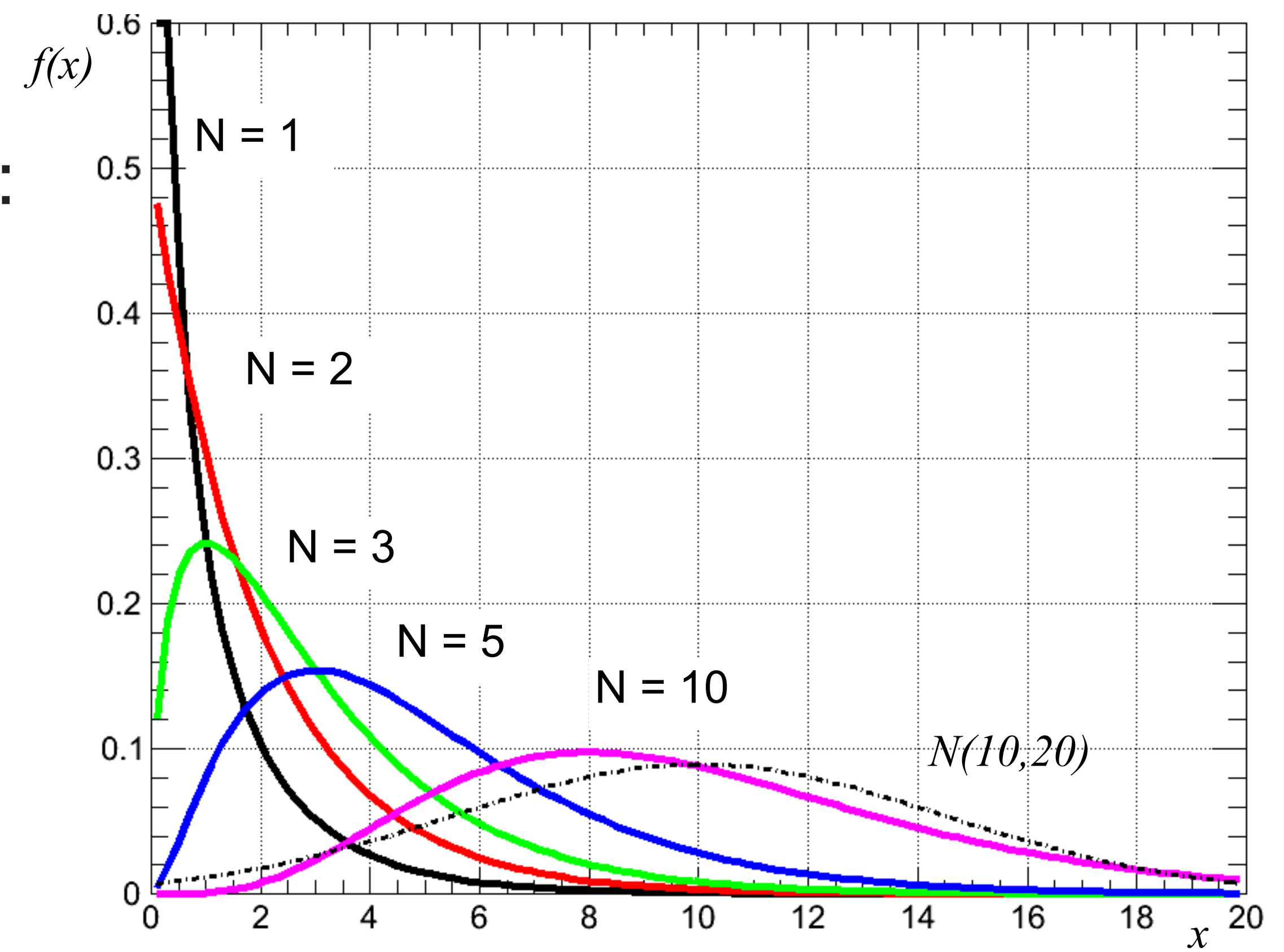
- If x_i are N independent, normally distributed random variables with mean 0 and variance, then the random variable $Q = \sum x_i^2$ is distributed according to the chi-square distribution with N degrees of freedom

$$f(x; N) = \frac{\frac{1}{2} \left(\frac{x}{2}\right)^{\frac{N}{2}-1} e^{-\frac{x}{2}}}{\Gamma\left(\frac{N}{2}\right)}$$

- Properties of the Chi-Square distribution:

- Mean: $\langle x \rangle = E(x) = N$

- Variance: $V(x) = 2N$



- Exponential probability density of the continuous variable $x > 0$:

$$N(x; \lambda) = \lambda e^{-\lambda x}$$

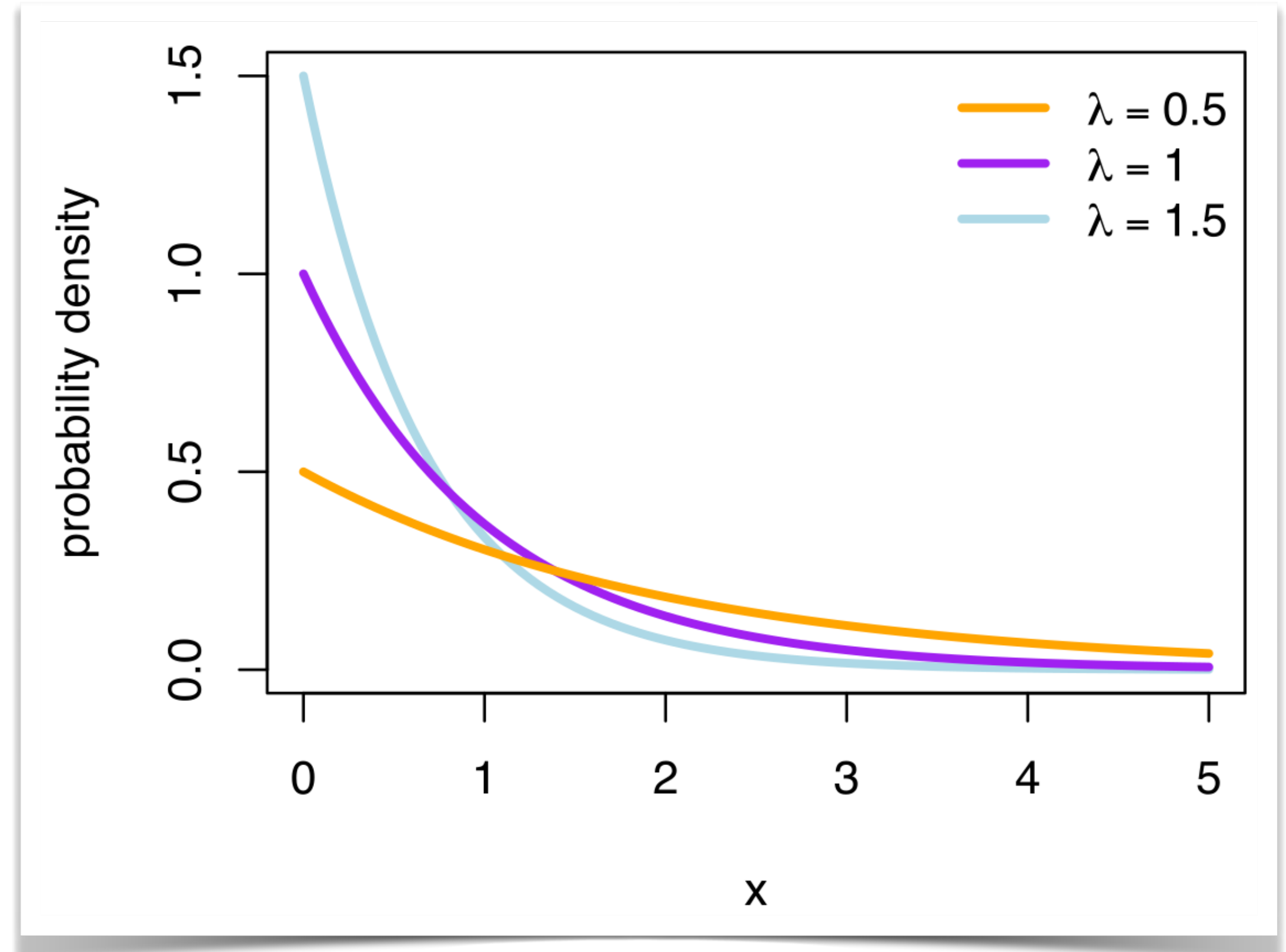
- Example: decay time of an unstable particle measured in its rest frame

- $\lambda = 1/\tau(\text{particle})$

- Properties of the Exponential distribution:

- Mean: $\langle x \rangle = E(x) = \frac{1}{\lambda}$

- Variance: $V(r) = \frac{1}{\lambda^2}$



- **Uniform** distribution

- Basic distribution for pseudo-random number generators

- **Gamma** distribution

- Probability model for waiting time

- **Cauchy** or **Lorentz** or **Breit-Wigner** distribution

- A solution to the differential equation describing a resonance
- Energy distribution of a resonance

- **Crystal Ball** distribution

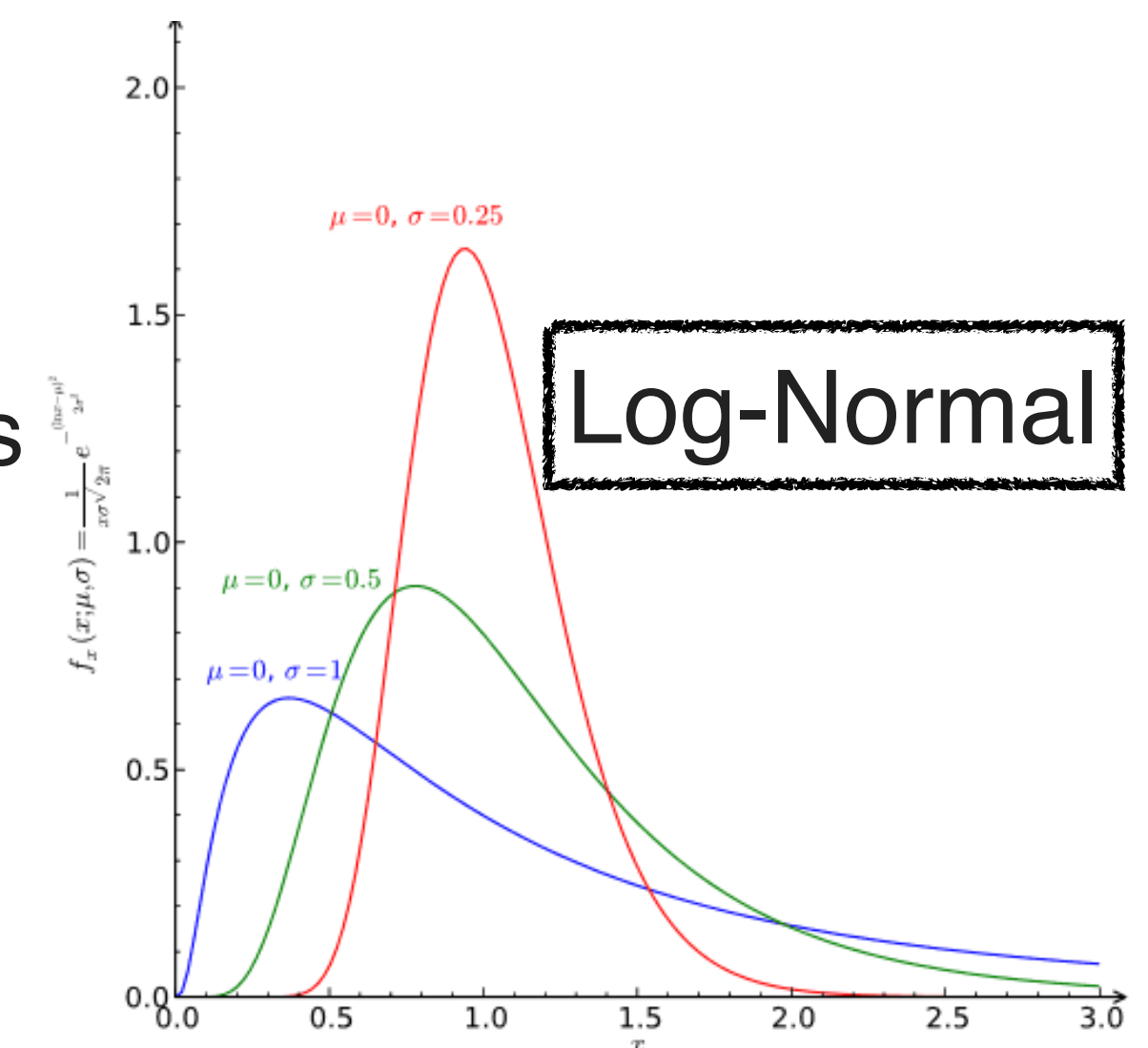
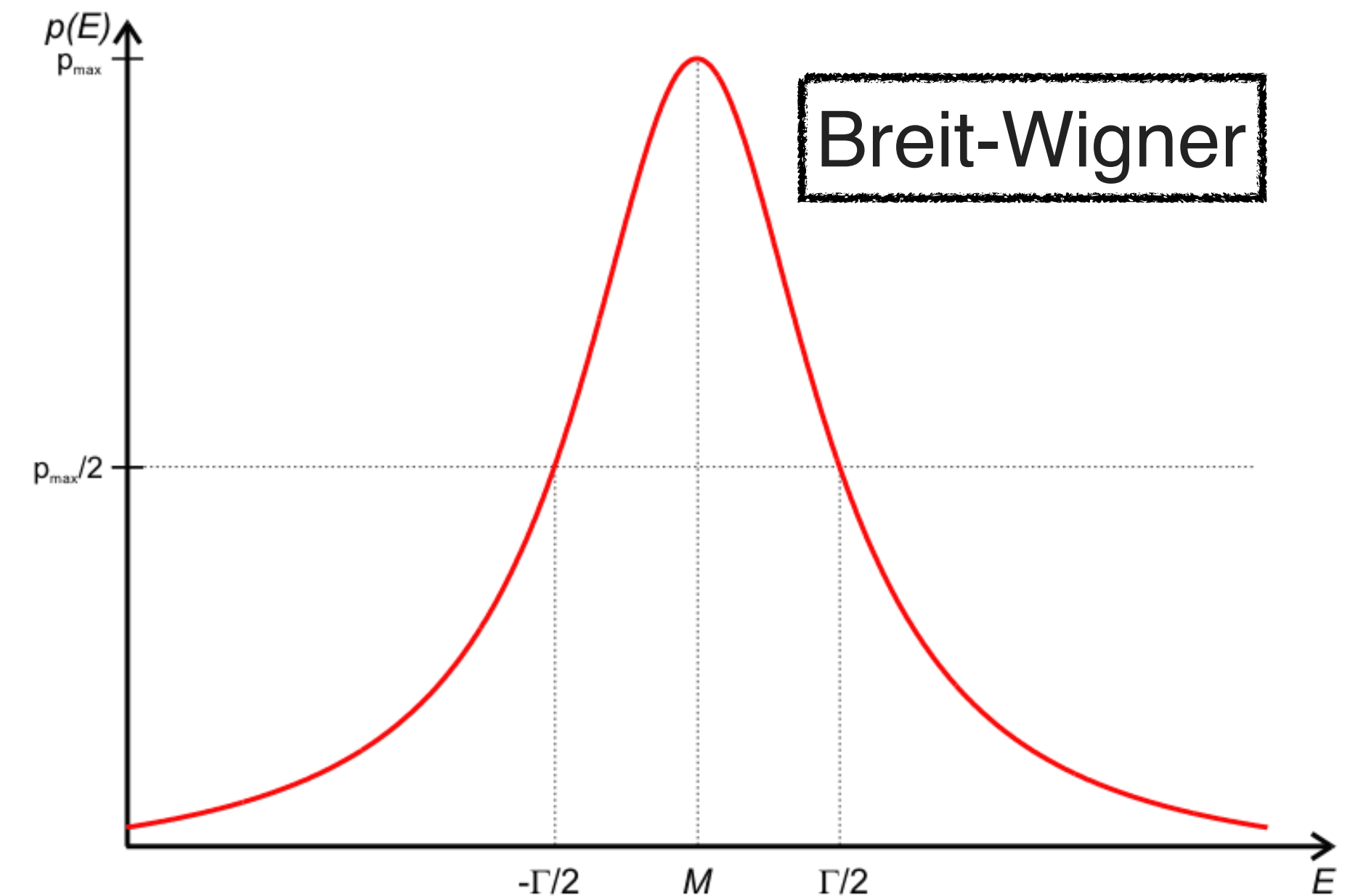
- Adds an asymmetric power-law tail to a Gaussian PDF

- **Landau** distribution

- Used to model the fluctuations in the energy loss of particles in thin layers

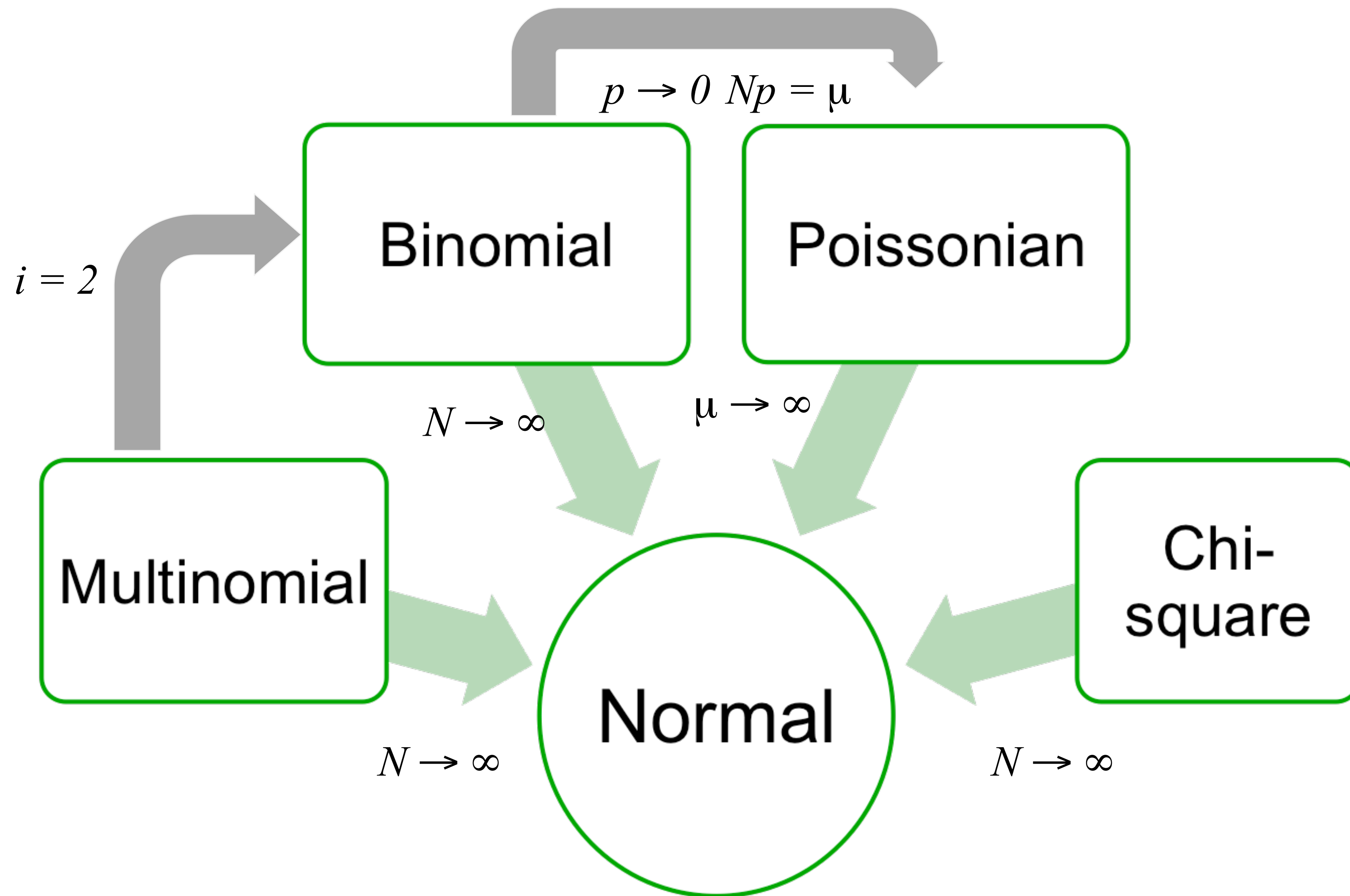
- **Log-Normal** distribution

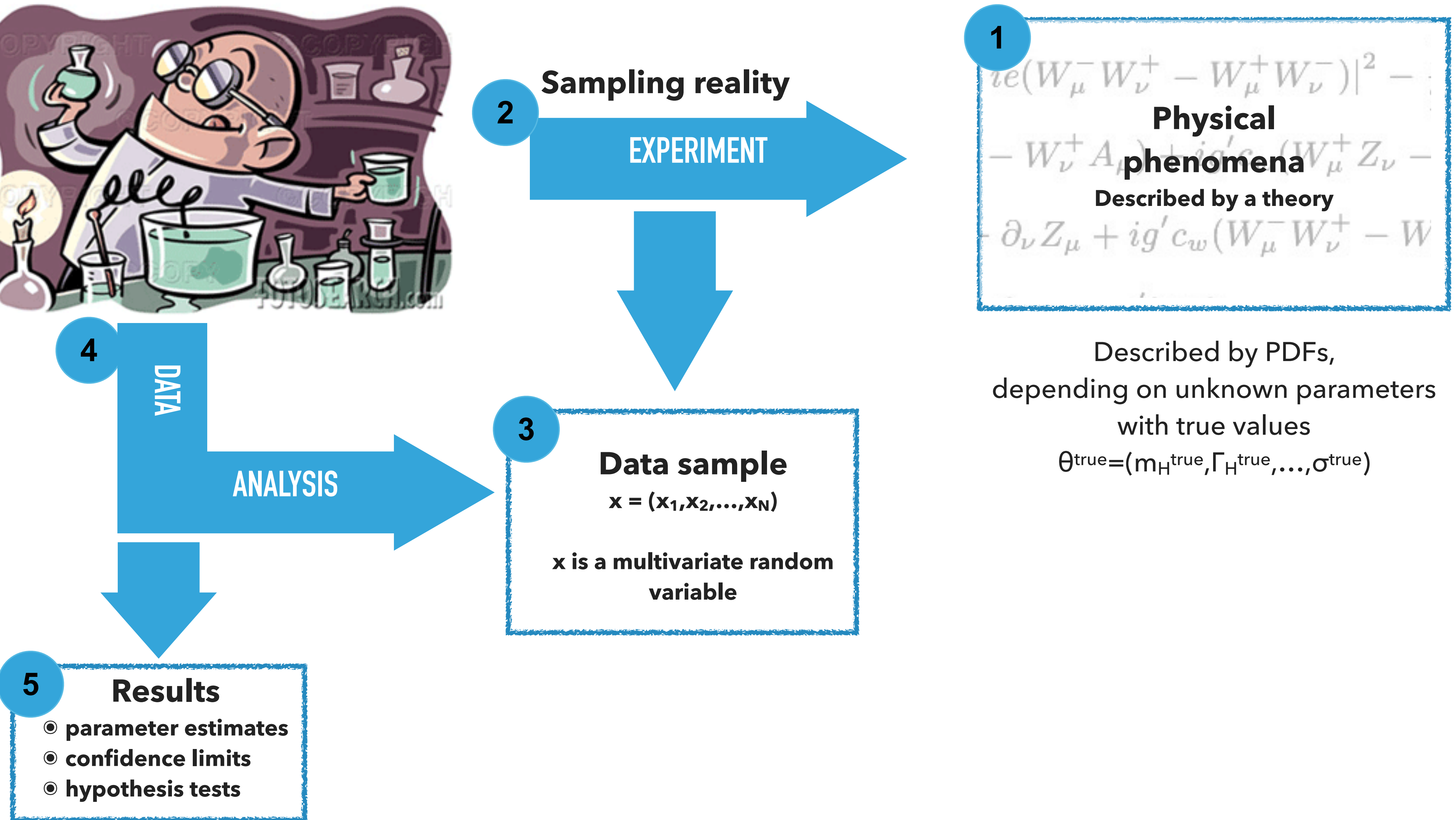
- Used when including systematic errors in the analysis
- If x is Log-Normally distributed, then $\log(x)$ is Normally distributed



ALL ROADS LEAD TO ROME

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MONTE CARLO METHODS

WHAT ARE MONTE CARLO METHODS?

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- a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results
- the underlying concept is to use randomness to solve problems that might be deterministic in principle
- mainly used in three problem classes:
 - optimisation
 - numerical integration
 - generating draws from a PDF
- invented in the late 1940s by physicists while he was working on nuclear weapons projects Los Alamos National Laboratory
- the name Monte Carlo, which refers to the Monte Carlo Casino in Monaco

IMPORTANCE OF MC IN SCIENCE

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EXPERIMENT

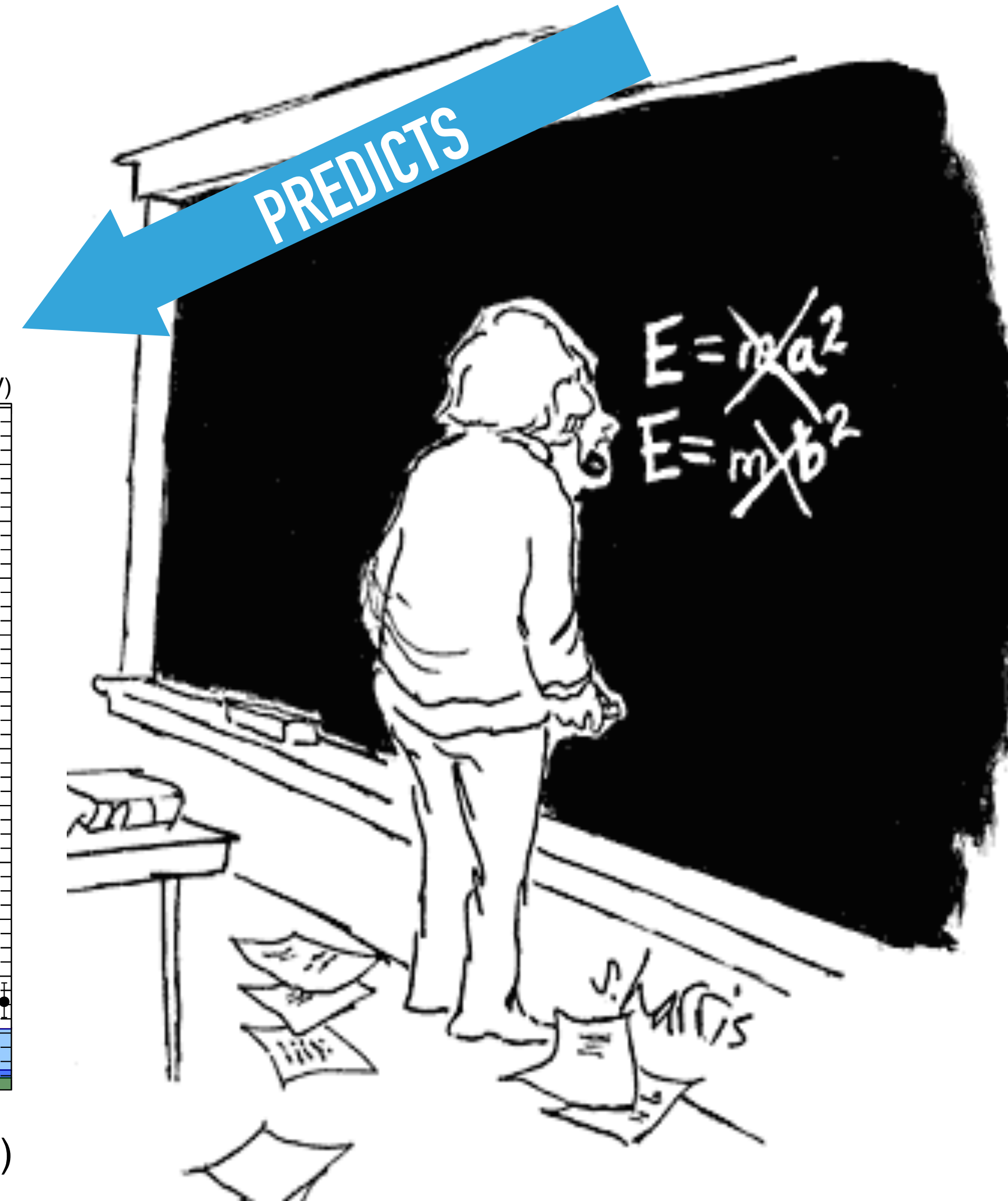
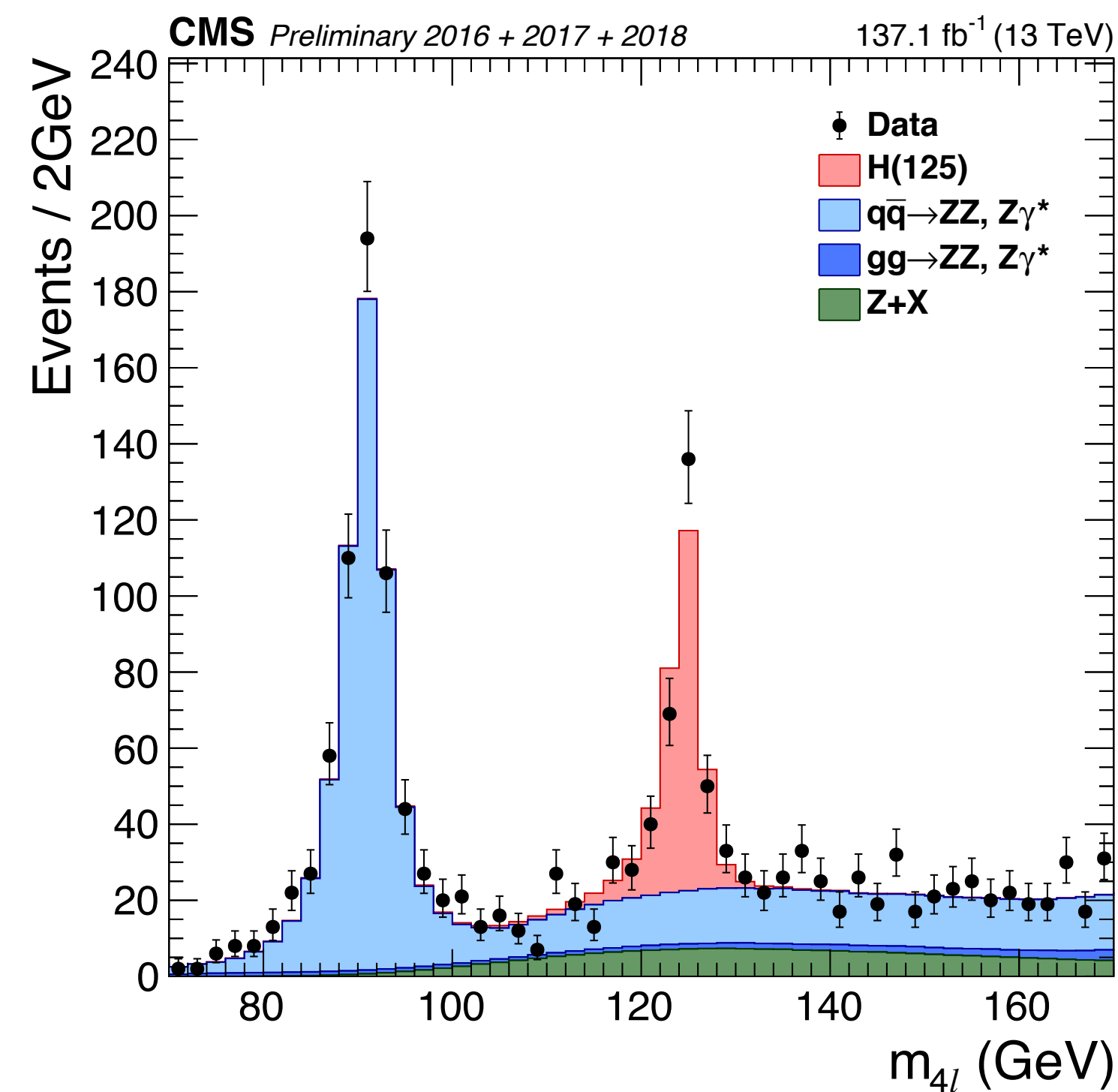
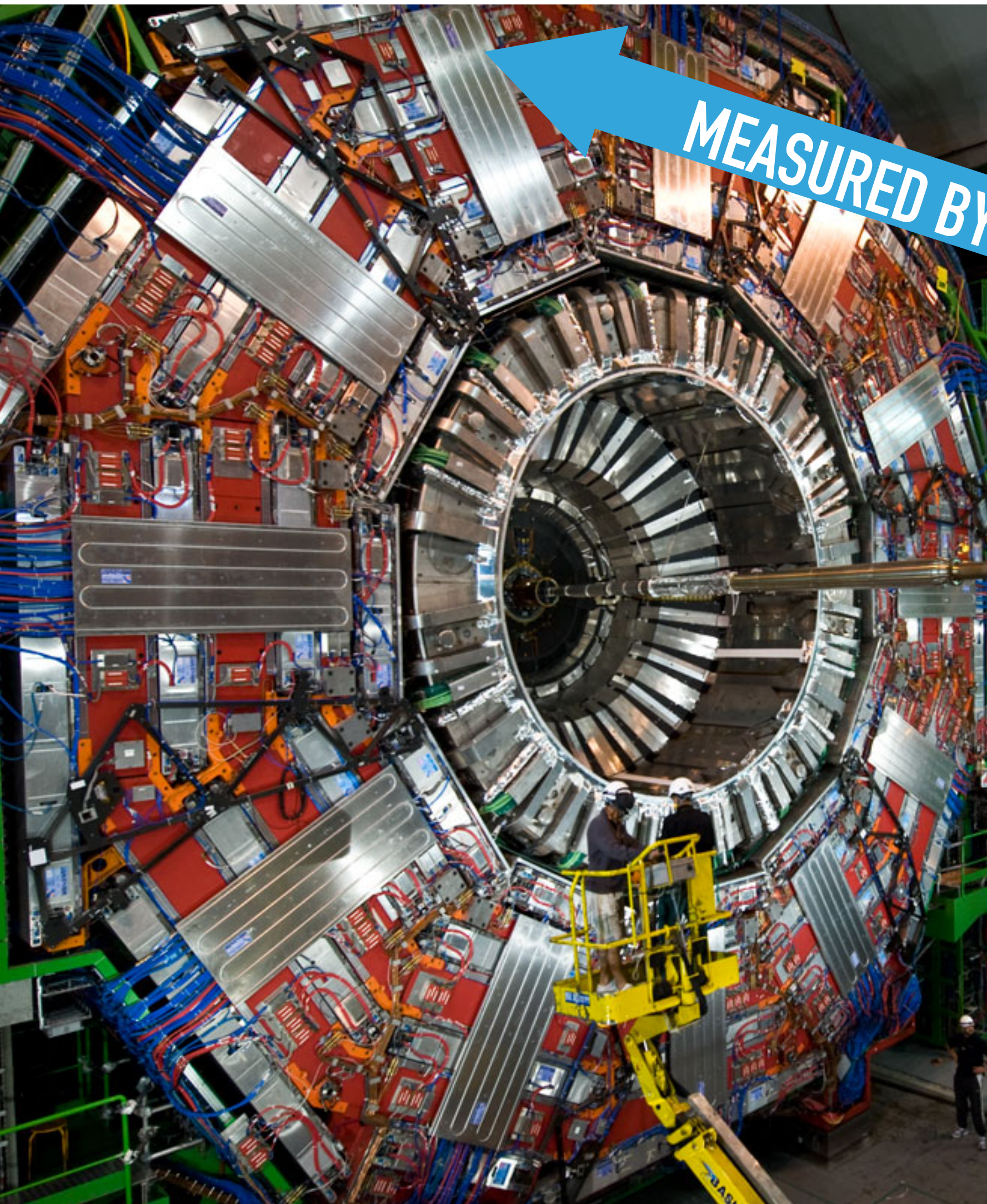
THEORY

AGREEMENT?

OBSERVABLES

PREDICTS

MEASURED BY



Monte Carlo Workflow

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DEFINE A DOMAIN OF POSSIBLE INPUTS



GENERATE INPUTS RANDOMLY FROM THE DOMAIN



PERFORM A DETERMINISTIC COMPUTATION USING THE INPUTS



AGGREGATE THE RESULTS OF THE INDIVIDUAL COMPUTATIONS INTO THE FINAL RESULT

ESTIMATING π

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DRAW A SQUARE ON THE GROUND, THEN INSCRIBE A CIRCLE WITHIN IT



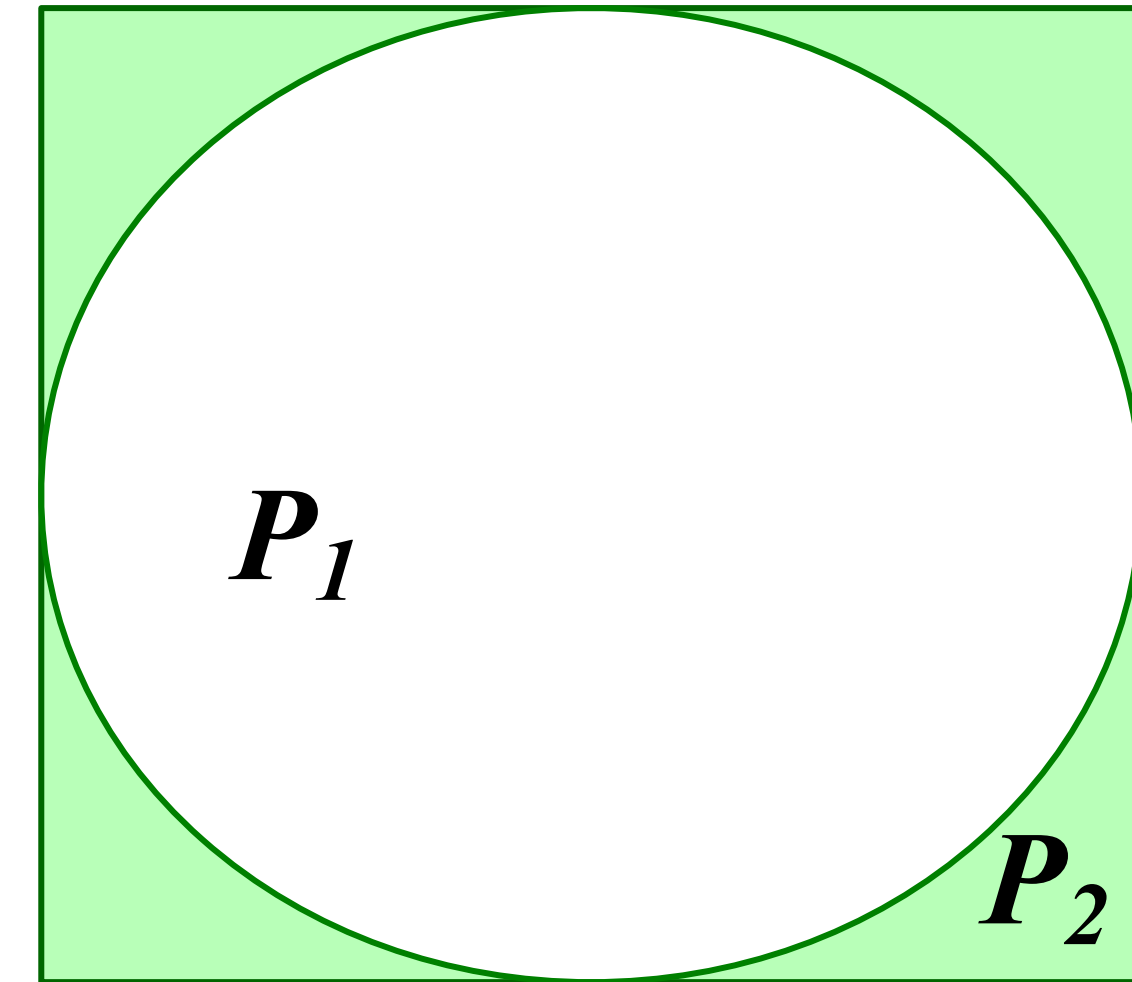
UNIFORMLY SCATTER N OBJECTS OF UNIFORM SIZE THROUGHOUT THE SQUARE.



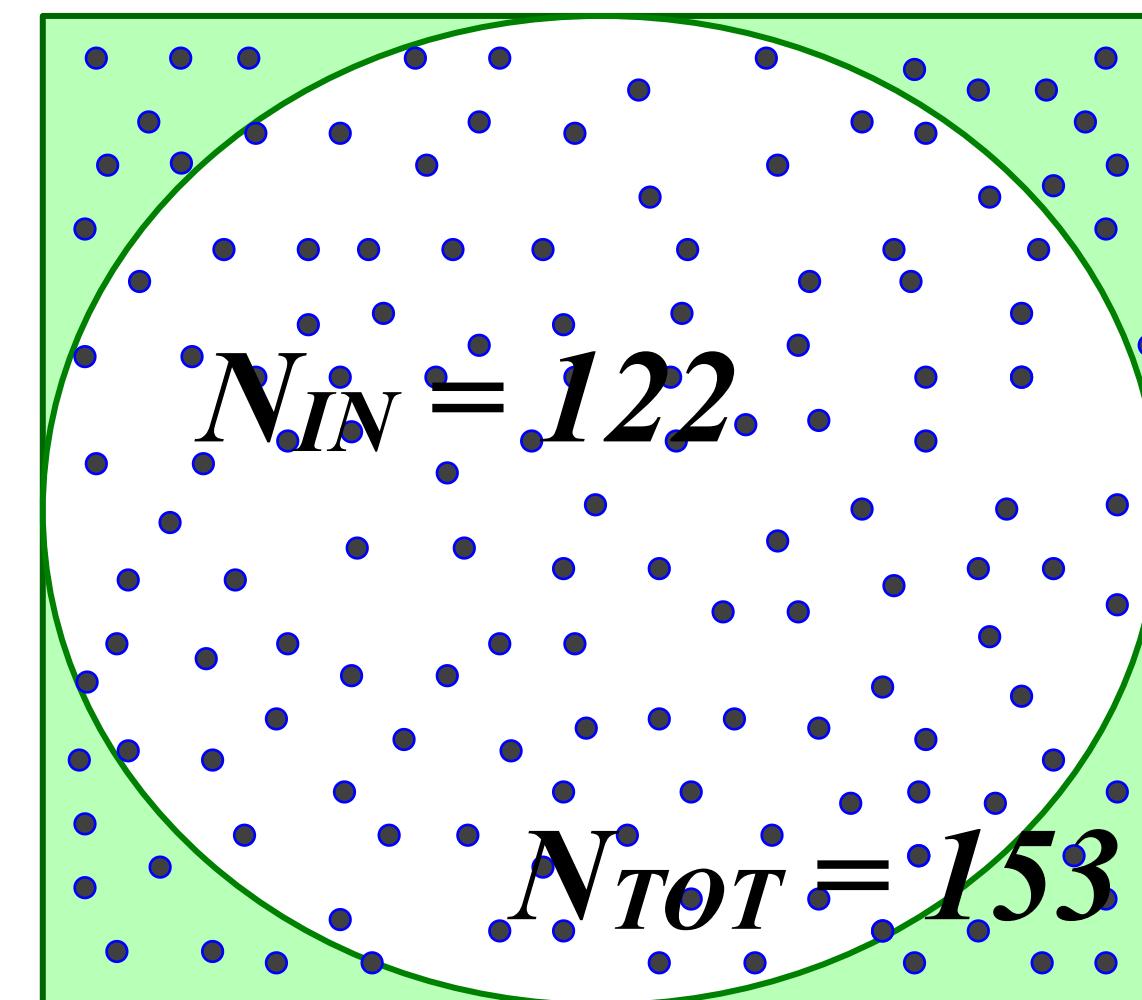
COUNT NUMBER OF OBJECTS IN THE CIRCLE = N_{IN}



ESTIMATE FINAL RESULT $\pi \sim 4 \times N_{IN} / N_{TOT}$

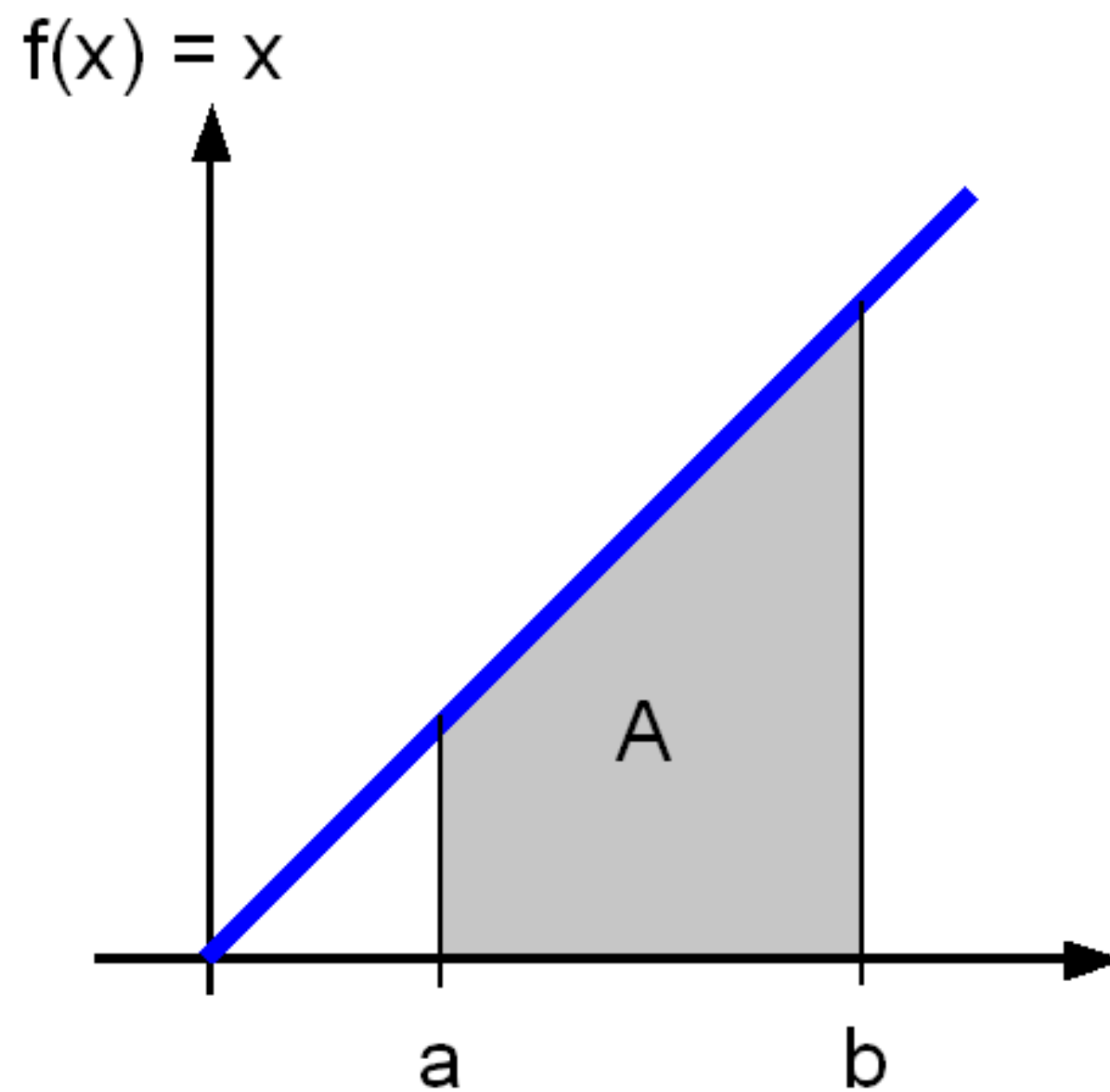


$$P_1 / P_2 = \pi / 4$$



$$\pi \approx 4 \times 122 / 153 = 3.19$$

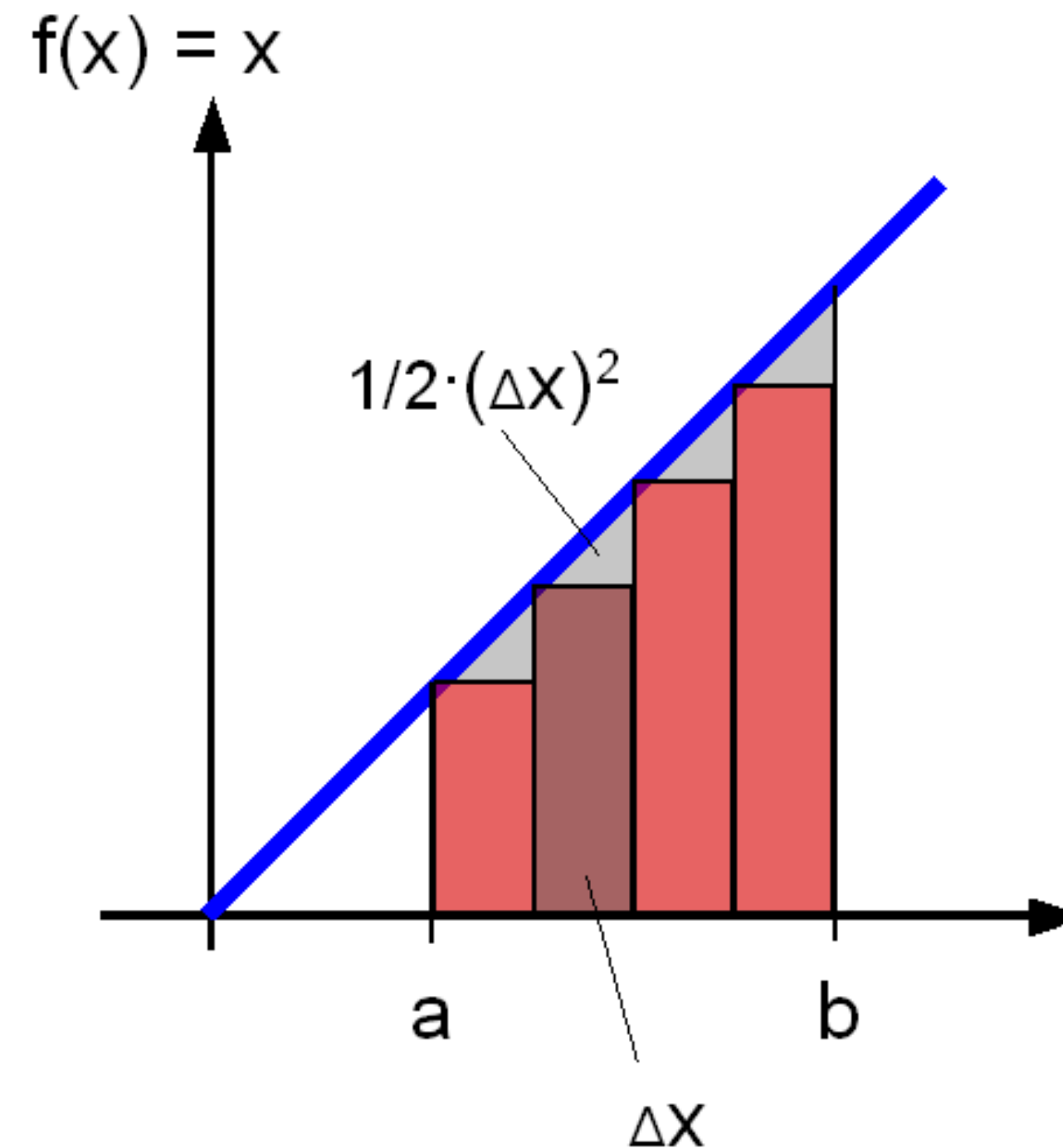
ANALYTICAL SOLUTION



$$\int x dx = \frac{1}{2}x^2$$

$$A = \frac{1}{2}(b^2 - a^2)$$

DETERMINISTIC ALGORITHM

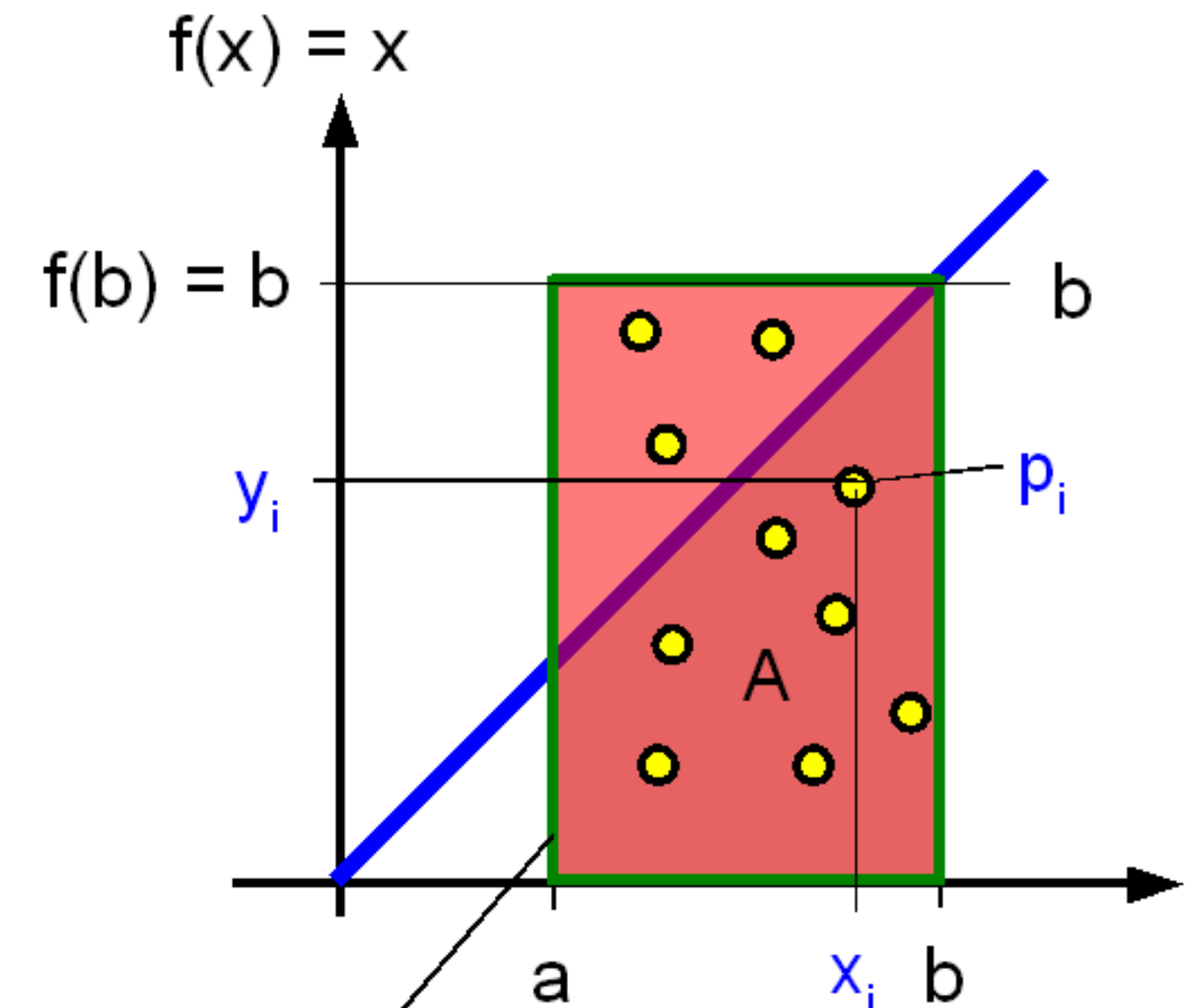


$$\Delta x = \frac{b - a}{n}$$

$$A \approx \sum f(a + i\Delta x)\Delta x$$

gets more precise with
more steps (n)

MONTE CARLO



$$B = (b-a) \cdot b$$

$$A \approx B \frac{N_{IN}}{N_{TOT}}$$

gets more precise with
more random number
pairs $p_i(x_i, y_i)$

- Physical methods:
 - “true” random numbers from “unpredictable” process
 - Example: dice, coin flipping, roulette
- True random numbers from random atomic or subatomic physical phenomena:
 - Example: radioactive decay, amplitude of noise in radio
- Computational methods:
 - Pseudo-random number generators create long runs (for example, millions of numbers long) with good random properties but eventually the sequence repeats
 - Example: Linear congruential generator



MC SIMULATION VS REAL LIFE

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EVENT GENERATION

TOOLS: MC GENERATORS (PYTHIA, ...)
OUTPUT: FINAL STATE PARTICLES



DETECTOR SIMULATION

TOOLS: MC SIMULATORS (GEANT)
OUTPUT: SIMULATED DETECTOR RESPONSE



EVENT RECONSTRUCTION

TOOLS: DETECTOR SOFTWARE PACKAGES (CUSTOM MADE; MC USED IN ALGORITHMS)
OUTPUT: RECONSTRUCTED PHYSICAL OBJECTS (ELECTRONS, MUONS, JETS ...)



DATA ANALYSIS

TOOLS: STATISTICS (ROOT, ...; MC USED IN ALGORITHMS; F.G. TOY MC)
OUTPUT: NEW KNOWLEDGE (PARAMETER/INTERVAL ESTIMATES, COST/PERFORMANCE ESTIMATE, HYPOTHESIS TESTS, ARTICLE, TALKS ...)

COLLISIONS

TOOLS: ACCELERATORS
OUTPUT: FINAL STATE PARTICLES



DATA ACQUISITION

TOOLS: DETECTORS (CMS, ATLAS,...)
OUTPUT: DETECTOR RESPONSE

